Probability, Context and Differentiation: The Layers

Steps 1–5: the distribution monad, the provenance comonad, the tangent layer, and orders of transitions
(working draft)

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October 2026

Abstract

The three companions [1, 2, 3] use three structures side by side: probability laws and kernels, provenance (which source a record came from), and differentiation at the boundary of the simplex. This paper studies how they fit together, one layer at a time. Every ingredient of Step 1 is standard in categorical probability; what this step contributes is a set of exact statements about where the ingredients *fail* to commute, which the later steps build on.

- The finite distribution monad $\mathcal{D}$ and the provenance comonad $W = L \times (-)$ are related by a distributive law $\lambda : W\mathcal{D} \Rightarrow \mathcal{D}W$, $(l, p) \mapsto \delta_l \otimes p$. All four axioms are written out with their types and proved. Its two-sided Kleisli category is the category of Markov kernels indexed by a source, with one source held throughout a composite.
- Marginalisation is total and affine. Event conditioning is partial and nonlinear. It is undefined exactly where the event has probability zero, it cannot be extended continuously to that wall, and its derivative grows like $1/p(A)$ near it. Conditioning commutes with marginalisation only for events that are cylinders over the kept coordinates.
- *Contextual Bayes decomposition.* Conditioning a pooled law equals pooling the conditioned sources with Bayes-updated weights. In the worked example, source weights $\frac{1}{4}, \frac{3}{4}$ become $\frac{8}{23}, \frac{15}{23}$, and keeping the old weights gives the wrong answer ($\frac{63}{160}$ instead of $\frac{9}{23}$).
- Forgetting provenance is marginalisation over the sources. It commutes with every window, but it is not injective: two families of sources can share a pooled law and every pooled table while differing in every labelled table.
- *Step 2, the tangent layer.* The cone tangent structure of the tangent companion commutes with the provenance comonad. Each source is a face of the labelled simplex, and a feasible direction leaves that face exactly when it injects whole records under another label; after forgetting provenance, an injected source and a drifting source have the same direction. Marginalisation and conditioning map feasible cones *onto* feasible cones, with explicit lifts; conditioning's lift shrinks like $p(A)$ near its wall. The cone $T^k H$ of the half-line is generated by elementary tensors, so it is closed under products, and pooling is a morphism at every order: the pooling paradox lives in the decision functional, not in pooling. Differentiating the contextual Bayes decomposition splits every sensitivity into a reweighting term and a within-source term; in the worked example the answer $\frac{9}{23}$ moves at rate $-\frac{16}{529}$ per unit of small-trial share.
- *Strata and jumps.* The sources that survive conditioning are exactly those with $\pi_l p_l(A) > 0$. Conditioning maps fibres of forgetting onto fibres: it keeps provenance ambiguity on the event, discards it off the event, and can destroy identifiability but never create it. Inside

its domain, conditioning moves at rate at most $2/P(A)$, so nothing jumps and a sudden change needs a rare event. At the wall the limit is the conditional of the path's leading jet; analytic passages are removable, and jumps need a kink, a non-analytic path, or a jump of the law.

- *Step 3, orders of transitions.* On non-negative germs in a Hardy field, order is a semiring homomorphism to $(\min, +)$, so every layer operation has an exact order calculus. Each source reaches an event at its own order. The sources of least order survive conditioning, with weights given by leading coefficients; the others vanish at their extinction orders. The two sides of a transition carry independent order data, analytic passages are removable but smooth ones need not be, and flat approaches have limits inside a Hardy field but not outside. A transition of order m is invisible to the tangent tower below level m. At a fixed gap, a higher-order approach switches more gradually.
- *Step 4a–4b, several parameters.* When several rates vanish together, orders form a fan: the normal fan of a Newton polyhedron, one cone per bounded face, each with its own surviving sources. The limit set is computed cone by cone. A limit exists when the exponents have a componentwise minimum, and for monomial cells only then. Lines through the limit point are not enough, and degenerate directions hide whole intervals. Iterated limits are the corners of the fan, and they are exactly what the tangent tower computes: the lexicographic lemma reads off one corner, and the canonical flip moves to the other. Five small examples show each case.
- *Step 4c, parameters as an object.* Parameters get their own 2-category, the Para construction over source-indexed kernels, whose 2-cells are reparametrisations; arcs are 2-cells from the germ line, and the real Puiseux field holds all arcs at once. Every layer operation commutes with reparametrisation. Decisions live on the order fan: along every arc the act is eventually constant, ties at the limit are broken at higher order, and the decision fan transports along monomial reparametrisations. In the worked example, two study designs are two 2-cells, each filling one cone, so each has a limit where the raw race has none.
- *Step 4d, the decorated fan and Theorem J.* The order fan carries a decoration, a facewise presheaf of initial forms, and only the pair (a probability-decorated order fan) determines the limits. Pooling's whole curvature is its exact cross term. With the sample size as a parameter, the lab's Theorem J becomes an inequality of orders: the band is the leading coefficient of the cross term on the one ray where exposure stays of order one, since exposure sees the share at half weight and impact at full weight. The same fan marks where Gaussian bands are valid, with a Poisson ray and an empirical wall beyond it. On the Poisson ray an exact discrete band replaces J's, and beyond it a bounded adversary owns the rare event.
- *Step 4e, graded classes.* The classes $\mathcal{M}_k$ of maps that keep feasibility at order k form a strictly descending chain of subcategories. Every subtraction-free map lies in all of them, so every layer operation does, conditioning included, at every order. Only subtraction (contrasts, as in estimators and decision gaps) breaks feasibility.
- *Step 4f, degenerate directions.* A direction needs refining only when the event's total mass $P(A)$ has a degenerate initial form, because non-negative cells cannot cancel. The resolution is Newton–Puiseux for $P(A)$: finite except along exact common roots, which lie in the wall. Two examples show a limit two refinements deep, and an infinite root cut by a competitor.
- *Step 5, functions, duality and logs.* States go forward and effects back. Conditioning, Bayes weights and Bayesian inversion are updates by effects, which compose by products, and the main paper's L^2 presheaf is the effect side of marginalisation. Under the log map, pooling tropicalises to min while Bayes stays exactly linear, so the order fan is the normal fan in log coordinates. Subtraction is the only operation without a log image. A cotangent cone at the boundary turns decisions into covector fields, and an exclusion principle collects what does not matter.

Thirteen exact checks (DL1–DL13) cover Step 1, fifteen (TL1–TL9, SJ1–SJ6) cover Step 2, seven (OR1–OR7) cover Step 3, seven (MP1–MP7) cover Step 4a–4b, six (PR1–PR6) cover Step 4c, seven (ST1–ST7) cover Step 4d, eleven (PB1–PB5, GC1–GC6) cover the Poisson band and Step 4e, six (DG1–DG6) cover Step 4f, and nine (FD1–FD9) cover Step 5.

1 Introduction and plan

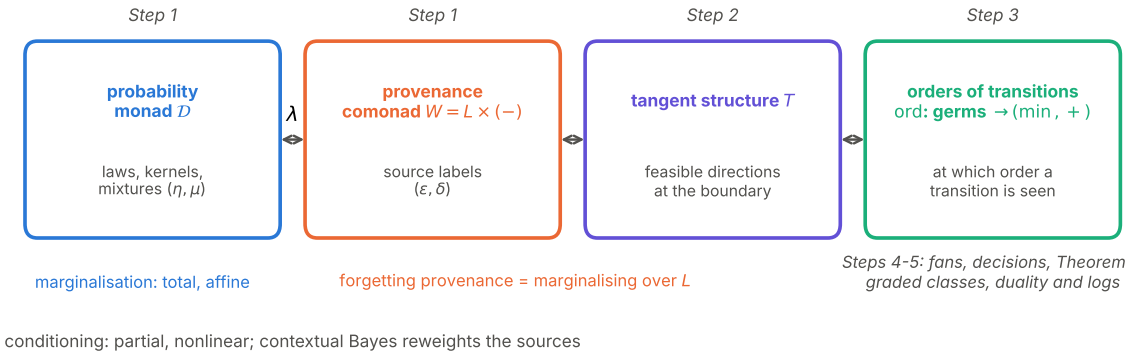


Figure 1: The layers and the step that builds each. Step 1 covers the probability monad, the provenance comonad and their distributive law λ, with marginalisation and conditioning. Step 2 brings in the tangent structure on simplices with boundary (section 8), Step 3 the orders of transitions (section 16), Step 4a–4b several parameters and the order fan (sections 17 and 18), Step 4c parameters as an object with decisions on the fan (section 19), and Step 4d the probability-decorated fan and Theorem J on a statistical fan (section 20). Step 4e collects the graded classes of maps (section 21), Step 4f resolves degenerate directions (section 22), and Step 5 adds functions, duality and logs (section 23).

What the layers are for. The lab [1] pools records from several sources, conditions on observed events, and asks how a decision moves when the data move. Each operation is classical on its own. What the programme keeps needing is the *compatibility* of these operations. Does conditioning commute with pooling? What does forgetting a source label lose? How do these operations behave at the boundary of the simplex, where the tangent companion [3] lives? The layers of fig. 1 answer these questions one at a time. Each section has four parts: a definition, a compatibility statement, a proof, and checks with their limits. All sets are finite.

A word on "context". In the contextuality literature, and in our covers and lab papers, a *context* is a measurement context: a window of jointly observed variables [4]. That is not the context of this paper. Here L is a set of *provenances*, the sources records come from, and $W = L \times (-)$ is the *provenance comonad* of the lab [1, Section "sources"]. The title keeps the broader word because later steps also carry measurement contexts.

2 Related work and what is new

Probability monads and Markov categories. The finite distribution monad is the finite-support part of the Giry monad [5]. Its Kleisli category FinStoch is the basic Markov category [6], in which joints, marginals and independence are handled by the bimonoidal structure of probability monads [7]. Effectuses give a related account of normalisation and conditioning [8].

Conditioning, disintegration and inversion. FinStoch has conditionals: every joint law factors through a kernel, which is unique almost surely and arbitrary on null events [6]. Bayesian inversion reverses a kernel relative to a prior [9]. Section 6 uses a different, partial operation, event conditioning normalised by $p(A)$, because the geometry of the later steps needs its behaviour as $p(A) \to 0$. Proposition 10 shows exactly how the two differ at the wall.

Context, sheaves and descent. Probability sheaves organise probability over a site of sample spaces [10], and the sheaf-theoretic account of contextuality organises it over measurement contexts [4]. Our covers paper works in that second setting. The provenance comonad is simpler: an explicit parameter carried alongside the law. Comonads of this product form are the environment comonads of comonadic semantics [11]. Distributive laws between a monad and a comonad, and the two-sided Kleisli category they produce, are classical [24, 12].

Differential structure. Tangent and differential categories axiomatise derivatives in general; the tangent companion builds a weakened tangent structure on simplices with boundary, with graded classes of admissible maps [3]. Step 2 (sections 8 to 13) connects that work to the layers here.

What is new. The distributive law of section 4 is the strength of a commutative monad, and its two-sided Kleisli category is the standard category of parameter-indexed kernels. We state and check them only to fix notation and types. The contribution of Step 1 is a set of exact statements about failure of commutation that the later steps use:

- conditioning a pooled law reweights the sources (theorem 11);
- forgetting provenance loses source contrasts that every pooled table misses (proposition 13);
- conditioning commutes with marginalisation only for cylinder events (proposition 8(4));
- event conditioning has no continuous extension to its wall (proposition 10).

The research gap is their compatibility with a provenance comonad, with higher-order feasible differentiation on finite simplices, and with decision functionals. In particular: how source reweighting, pooling and decisions interact with the boundary geometry and with the graded classes.

3 The finite distribution monad

Definition 1. For a finite set X, $\mathcal{D}X$ is the set of probability vectors on X. For $f : X \to Y$, $\mathcal{D}f$ is the push-forward. The unit is $\eta_X(x) = \delta_x$. The multiplication $\mu_X : \mathcal{D}\mathcal{D}X \to \mathcal{D}X$ flattens a finitely supported distribution of distributions: $\mu_X(\Phi)(x) = \sum_p \Phi(p)\, p(x)$.

In the lab, pooling labelled batches is μ. The Kleisli category is FinStoch, the Markov category of [1, Section "Markov categories"].

4 The provenance comonad and the distributive law

Definition 2 (provenance)**.** For a finite set L of sources, $WX = L \times X$, with counit $\epsilon_X(l, x) = x$ (forget the source) and comultiplication $\delta^W_X(l, x) = (l, (l, x))$ (copy it). The distributive law is

$$\lambda_X : W\mathcal{D}X \longrightarrow \mathcal{D}WX, \qquad \lambda_X(l, p) = \delta_l \otimes p,$$

the law of (l, x) with l fixed and $x \sim p$.

A comonad W distributes over a monad T through a transformation $WT \Rightarrow TW$ satisfying four axioms [12]. The direction matters: λ has the comonad outside on its source and the monad outside on its target.

Proposition 3 (the four axioms)**.** *λ is a distributive law of the comonad W over the monad $\mathcal{D}$. Explicitly, the following commute for every finite X.*

(1) Unit. $\lambda_X \circ W\eta_X = \eta_{WX} : \ WX \to \mathcal{D}WX.$

(2) Multiplication. $\lambda_X \circ W\mu_X = \mu_{WX} \circ \mathcal{D}\lambda_X \circ \lambda_{\mathcal{D}X} : \ W\mathcal{D}\mathcal{D}X \to \mathcal{D}WX.$

(3) Counit. $\mathcal{D}\epsilon_X \circ \lambda_X = \epsilon_{\mathcal{D}X} : \ W\mathcal{D}X \to \mathcal{D}X.$

(4) Comultiplication. $\mathcal{D}\delta^W_X \circ \lambda_X = \lambda_{WX} \circ W\lambda_X \circ \delta^W_{\mathcal{D}X} : \ W\mathcal{D}X \to \mathcal{D}WWX.$

$$\begin{array}{ccccc} W\mathcal{D}\mathcal{D}X & \xrightarrow{\lambda_{\mathcal{D}X}} & \mathcal{D}W\mathcal{D}X & \xrightarrow{\mathcal{D}\lambda_X} & \mathcal{D}\mathcal{D}WX \\ {\scriptstyle W\mu_X}\downarrow & & & & \downarrow{\scriptstyle \mu_{WX}} \\ W\mathcal{D}X & & \xrightarrow[\lambda_X]{} & & \mathcal{D}WX \end{array} \qquad \begin{array}{ccccc} W\mathcal{D}X & \xrightarrow{\delta^W_{\mathcal{D}X}} & WW\mathcal{D}X & \xrightarrow{W\lambda_X} & W\mathcal{D}WX \\ {\scriptstyle \lambda_X}\downarrow & & & & \downarrow{\scriptstyle \lambda_{WX}} \\ \mathcal{D}WX & & \xrightarrow[\mathcal{D}\delta^W_X]{} & & \mathcal{D}WWX \end{array}$$

Proof. Evaluate both sides on an element.

- (1) $(l, x) \mapsto \lambda(l, \delta_x) = \delta_l \otimes \delta_x = \delta_{(l,x)}$.
- (2) For (l, Φ) the left side is $\delta_l \otimes \mu\Phi$. On the right, $\lambda_{\mathcal{D}X}(l, \Phi)$ is the law of (l, p) with $p \sim \Phi$; $\mathcal{D}\lambda$ sends it to the law of $\delta_l \otimes p$; μ flattens it to $\sum_p \Phi(p)\, \delta_l \otimes p = \delta_l \otimes \mu\Phi$.
- (3) Pushing $\delta_l \otimes p$ forward along $(l, x) \mapsto x$ gives $p = \epsilon_{\mathcal{D}X}(l, p)$.
- (4) The left side is the law of $(l, (l, x))$ with $x \sim p$. On the right, $(l, p) \mapsto (l, (l, p)) \mapsto (l, \delta_l \otimes p) \mapsto \delta_l \otimes (\delta_l \otimes p)$, which is the same law.

Checks DL1–DL4 confirm each identity exactly, in rational arithmetic. □

Proposition 4 (source-indexed Markov kernels)**.** *The two-sided Kleisli category of λ [24, 12] has the finite sets as objects. A morphism $X \to Y$ is a map $f : L \times X \to \mathcal{D}Y$, a Markov kernel indexed by the source. Composition is $g \circ f = \mu \circ \mathcal{D}g \circ \lambda \circ Wf \circ \delta^W$, which is*

$$(g \circ f)(l, x)(z) = \sum_y f(l, x)(y)\, g(l, y)(z),$$

and the identity is $\eta \circ \epsilon$. One source is held throughout a composite.

Proof. δ^W copies l; Wf uses one copy to produce $f(l, x)$; λ pairs the other copy with it; $\mathcal{D}g$ applies $g(l, \cdot)$; μ sums. Associativity and the identity laws follow from proposition 3; DL5 checks them directly. □

Limits. Sources here are known and discrete. A source drawn at random is a law on $L \times X$, an element of $\mathcal{D}WX$, and is handled through λ and μ; a source that is uncertain inside a kernel is not.

5 Marginalisation: total and affine

Definition 5. For $X = \prod_{i \in V} X_i$ and $S \subseteq V$, the marginalisation $M_S = \mathcal{D}\pi_S : \mathcal{D}X \to \mathcal{D}X_S$ is the push-forward along the projection.

Proposition 6 (marginalisation)**.** *(1) M_S is total, and it preserves positivity and normalisation.*

(2) $M_S \circ M_T = M_S$ for $S \subseteq T$, because $\pi_S \circ \pi_T = \pi_S$ and $\mathcal{D}$ is a functor.

(3) M_S is affine (linear on measures), so $D^2 M_S = 0$.

(4) It commutes with forgetting provenance: $M_S \circ \mathcal{D}\epsilon = \mathcal{D}\epsilon \circ \mathcal{D}(W\pi_S)$ on $\mathcal{D}WX$.

Proof. (1)–(3) are immediate from the push-forward. (4) $\pi_S \circ \epsilon = \epsilon \circ W\pi_S$ as maps $L \times X \to X_S$, and $\mathcal{D}$ is a functor (DL6, DL11). □

6 Conditioning: partial and nonlinear

Definition 7. For an event $A \subseteq X$, the conditioning operator is the partial map

$$C_A : \mathcal{D}X \rightharpoonup \mathcal{D}X, \qquad C_A(p) = \frac{p\,\mathbf{1}_A}{p(A)}, \qquad \operatorname{Dom} C_A = \{p : p(A) > 0\}.$$

For $Y : X \to Z$ and $z \in Z$, write $C_z = C_{\{Y=z\}}$.

Proposition 8 (conditioning)**.** *(1)* Partiality. *C_A is defined exactly on $\{p(A) > 0\}$, a relatively open subset of the simplex whose boundary is the face $\{p(A) = 0\}$, the* wall.

(2) Total probability. $p = \sum_{z : p(Y=z) > 0} p(Y{=}z)\, C_z(p)$.

(3) Iteration. $C_B(C_A(p)) = C_{A \cap B}(p)$ *whenever* $p(A \cap B) > 0$.

(4) With marginalisation. *If $A = \pi_S^{-1}(A')$ is a cylinder over the kept coordinates, then $M_S \circ C_A = C_{A'} \circ M_S$. For other events the square does not commute in general.*

(5) Derivatives. *Write N_A for multiplication by $\mathbf{1}_A$ and $a(h) = h(A)$. Then*

$$DC_A(p)[h] = \frac{N_A h}{a(p)} - \frac{N_A p\, a(h)}{a(p)^2}, \qquad D^2 C_A(p)[h,k] = -\frac{N_A h\, a(k) + N_A k\, a(h)}{a(p)^2} + \frac{2 N_A p\, a(h) a(k)}{a(p)^3}.$$

So conditioning is nonlinear, and $\|DC_A\|$ grows like $1/p(A)$ as p approaches the wall.

Proof. (1) The denominator. (2) $\sum_z p(Y{=}z)\, p\mathbf{1}_{\{Y=z\}}/p(Y{=}z) = p$. (3) $(p\mathbf{1}_A/p(A))\mathbf{1}_B/(p(A \cap B)/p(A)) = p\mathbf{1}_{A\cap B}/p(A \cap B)$. (4) With $A = \pi_S^{-1}(A')$, $M_S(p\mathbf{1}_A) = (M_S p)\mathbf{1}_{A'}$ and $p(A) = M_S p(A')$. (5) Differentiate the quotient (DL7, DL8, DL11). □

Example 9 (the square fails for a non-cylinder event)**.** Take two binary variables, R (recovered) and H (high risk), and the law p with $p(R{=}1, H{=}1) = \frac{9}{40}$, $p(R{=}1, H{=}0) = \frac{7}{20}$, $p(R{=}0, H{=}1) = \frac{1}{4}$ and $p(R{=}0, H{=}0) = \frac{7}{40}$ (the pooled law of example 12). Conditioning on $A = \{H = 1\}$ and then keeping R gives $P(R{=}1) = \frac{9}{19}$. The event projects to all of R, so conditioning the marginal on its projection changes nothing: $P(R{=}1) = \frac{23}{40}$. The event reweights R through the coordinate that was dropped (DL13).

Proposition 10 (conditioning at the wall)**.** *Let A have at least two points, and let p_0 satisfy $p_0(A) = 0$.*

(1) C_A has no continuous extension to p_0. Along the ray $p_t = (1-t)p_0 + t\,q$, with q supported in A, $C_A(p_t) = q$ for every $t > 0$. So every law on A is a limit at p_0.

(2) A conditional in the sense of FinStoch *[6] exists at p_0: any kernel works on the null event A, and every choice gives the same joint law. The partial map C_A and the categorical conditional therefore agree off the wall and part company exactly on it. There the conditional is defined by fiat, and no choice of it is continuous.*

Proof. (1) $p_t \mathbf{1}_A = t\,q$ and $p_t(A) = t$. (2) Conditionals in FinStoch are unique only almost surely; on a null event they are unconstrained. DL12 checks (1) exactly on three rays. □

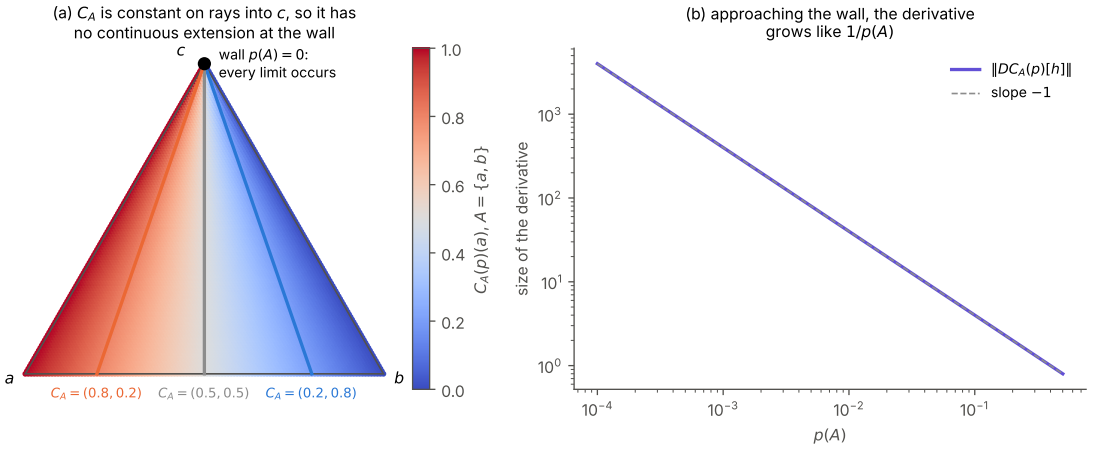


Figure 2: Conditioning near the wall. (a) On the triangle of laws on $\{a, b, c\}$ with $A = \{a, b\}$, the colour is $C_A(p)(a)$. It is constant along every ray into the vertex c, where $p(A) = 0$, so every value is a limit there (proposition 10). (b) The size of the derivative of C_A along a fixed direction, on log scales: slope -1, so it grows like $1/p(A)$ (proposition 8(5)).

This is why the later steps use the partial map. Its domain has a boundary, and the boundary carries the geometry. The categorical conditional hides the wall by assigning arbitrary values there. The tangent companion already places C_A in its second-order class of maps [3, Corollary "the second-order subcategory"]; Steps 2 and 3 take up corners and higher orders.

7 Provenance with marginalisation and conditioning

A pooled law is $\sum_l \pi_l p_l = \mathcal{D}\epsilon\big(\sum_l \pi_l\, \lambda(l, p_l)\big)$: label each source's law, mix, and forget the label.

Theorem 11 (contextual Bayes decomposition)**.** *Let $p_l \in \mathcal{D}X$ and weights π_l, and let A be an event with $\sum_j \pi_j p_j(A) > 0$. Then*

$$C_A\Big(\sum_l \pi_l p_l\Big) = \sum_{l:\, p_l(A) > 0} \pi_l^A\, C_A(p_l), \qquad \pi_l^A = \frac{\pi_l p_l(A)}{\sum_j \pi_j p_j(A)}.$$

Equivalently, conditioning the labelled law $\sum_l \pi_l\, \lambda(l, p_l)$ on $L \times A$ gives $\sum_l \pi_l^A\, \lambda(l, C_A(p_l))$. A known source stays known ($C_{L\times A}(\delta_l \otimes p) = \delta_l \otimes C_A(p)$), and the source weights are updated by Bayes' rule. Conditioning within each source and pooling with the old *weights gives the same law when $p_l(A)$ is the same for every source, and a different law in general.*

Proof. $C_A(\sum_l \pi_l p_l) = \sum_l \pi_l p_l \mathbf{1}_A / \sum_j \pi_j p_j(A)$, and $\pi_l p_l \mathbf{1}_A = \pi_l p_l(A)\, C_A(p_l)$. The labelled form is the same computation on $L \times X$. If all $p_l(A)$ are equal then $\pi^A = \pi$. If the $C_A(p_l)$ are affinely independent, the two pools agree only if $\pi^A = \pi$ (DL9: exact; the old weights were wrong in all 40 random cases). □

Example 12 (two sources)**.** Records come from small trials (source 1, weight $\frac{1}{4}$) and from a registry (source 2, weight $\frac{3}{4}$). Each records R (recovered) and H (high risk):

	$R{=}1, H{=}1$	$R{=}1, H{=}0$	$R{=}0, H{=}1$	$R{=}0, H{=}0$
source 1	3/10	1/2	1/10	1/10
source 2	1/5	3/10	3/10	1/5
pooled	9/40	7/20	1/4	7/40

Condition on recovery, $A = \{R = 1\}$. Recovery is likelier in source 1 ($\frac{4}{5}$ against $\frac{1}{2}$), so the recovered records come from it more often than its share: $\pi^A = (\frac{8}{23}, \frac{15}{23})$. Among the recovered, the high-risk share is $\frac{3}{8}$ in source 1 and $\frac{2}{5}$ in source 2. Pooled with the Bayes weights, it is $\frac{8}{23} \cdot \frac{3}{8} + \frac{15}{23} \cdot \frac{2}{5} = \frac{9}{23}$, which is C_A of the pooled law. Pooled with the old weights it would be $\frac{1}{4} \cdot \frac{3}{8} + \frac{3}{4} \cdot \frac{2}{5} = \frac{63}{160}$, which is wrong (fig. 3; DL13).

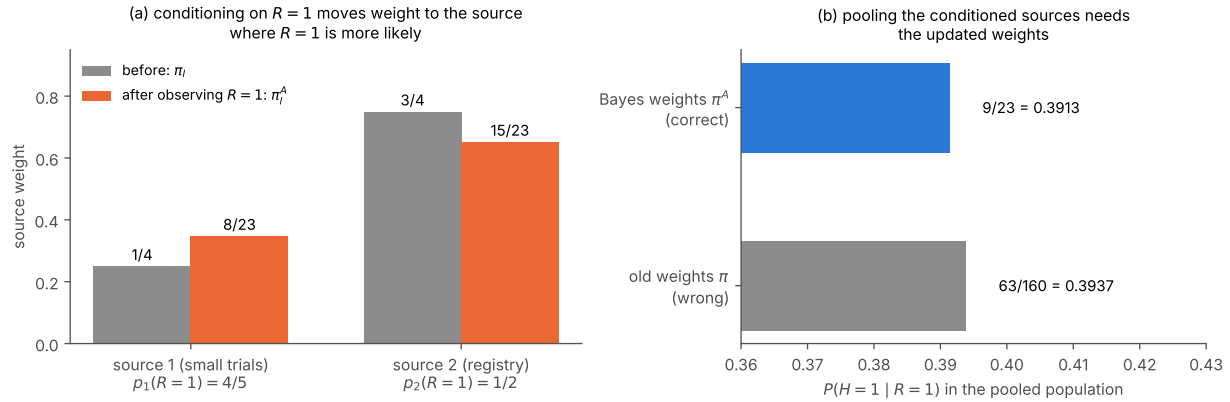


Figure 3: Example 12. (a) Conditioning on $R = 1$ moves weight towards the source in which recovery is likelier. (b) The high-risk share among the recovered, pooled with the updated weights ($\frac{9}{23}$, correct) and with the old ones ($\frac{63}{160}$, wrong).

Proposition 13 (forgetting provenance)**.** *Let $\Lambda = \sum_l \pi_l\, \lambda(l, p_l) \in \mathcal{D}WX$ be a labelled family, and $\mathcal{D}\epsilon(\Lambda) = \sum_l \pi_l p_l$ its pooled law.*

(1) Forgetting is marginalisation over L. It commutes with every window marginalisation: the pooled window tables are the window tables of the pooled law.

(2) Forgetting is not injective. For any law P and any direction Q with $\sum Q = 0$ and $P \pm Q \geq 0$, the families $\frac{1}{2}\lambda(l_1, P) + \frac{1}{2}\lambda(l_2, P)$ and $\frac{1}{2}\lambda(l_1, P+Q) + \frac{1}{2}\lambda(l_2, P-Q)$ have the same pooled law

and the same pooled window tables. Their labelled window tables differ on every window S with $M_S Q \neq 0$, by the source contrast $\pm M_S Q$.

(3) Forgetting does not commute with conditioning source by source: by theorem 11, conditioning reweights the sources.

Proof. (1) is proposition 6(4). (2) The pooled law is P in both cases, and the labelled tables are $M_S(P \pm Q)$ for the two labels. (3) is theorem 11 (DL10). □

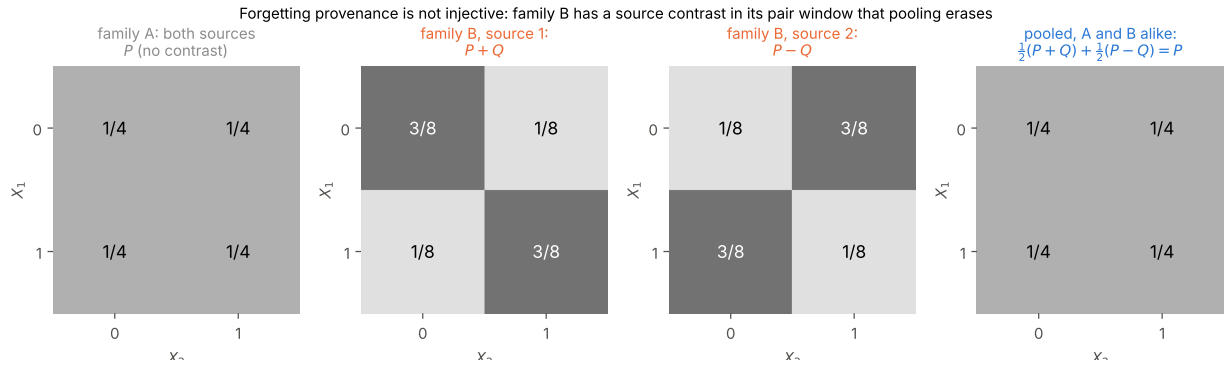


Figure 4: Proposition 13(2) on two binary variables: P uniform and $Q = \frac{1}{8}$ times the agreement contrast. Family A has no source contrast; family B has opposite contrasts in its two sources. Both pool to P, so no pooled table can tell them apart; the labelled tables can.

So a labelled cover sees source contrasts that pooling erases (fig. 4). In the lab these contrasts are the degree-zero discrepancy $\delta^0 m_T$ of the labelled descent tower and the source statistic of Theorem J [1, Theorem "sources and bands"]. Proposition 13 says only that forgetting loses them. How large a contrast pooling can hide, in standard errors, is Theorem J's band, a statistical statement derived in Step 4d (theorem 53).

8 Step 2: the tangent layer

Step 2 brings the cone tangent structure of the tangent companion [3] into the layers and asks how each Step 1 operation acts on feasible directions. Feasible directions matter because they are the ways data can move. In the lab, an adversary who adds a share ε of whole records from a law Q moves the law p to $(1-\varepsilon)p + \varepsilon Q$, that is, along $Q - p$ [1, Part II]. Every such direction is feasible, and every feasible direction is a positive multiple of one of them.

What we use from the tangent companion. For a law p on a finite set, the *feasible cone* is

$$C_p = \Big\{ u : \ \textstyle\sum_x u(x) = 0, \ u(x) \geq 0 \text{ wherever } p(x) = 0 \Big\},$$

the directions u for which $p + su$ is still a law for small $s \geq 0$. It is a vector space exactly when p has full support; at the boundary it is only a cone, because mass can be added to an empty cell but not removed from it. $T\Delta = \{(p, u) : u \in C_p\}$, and a map f acts by $Tf(p, u) = (f(p), Df(p)u)$. Higher orders are written in the Weil algebra $W_k = \mathbb{R}[e_1, \ldots, e_k]/(e_1^2, \ldots, e_k^2)$: a point of $T^2\Delta$ is $p + u\,e_1 + v\,e_2 + w\,e_1 e_2$, and $T^2 f$ is f evaluated on such points. Membership is decided cell by cell by the lexicographic lemma [3, Lemma "the lexicographic lemma"]. A map is a *morphism at*

order k if $T^k f$ maps T^k of its domain into T^k of its codomain. Stochastic matrices are morphisms at every order. A smooth map is a morphism at order 2 iff it passes the *second-order corner criterion*: at each output cell y with $f_y(p) = 0$, the Hessian of f_y is non-negative on pairs from $K_y = C_p \cap \ker Df_y(p)$ [3, Theorem "second-order cone morphisms"]. The criterion is not automatic: $q \mapsto \big(1 - (q_1 - q_2)^2, (q_1 - q_2)^2, 0\big)$ keeps first-order feasibility and fails at second order.

Pooling has several simplices as inputs (the weights and each source), so we need the criterion on a product.

Lemma 14 (the criterion on products of simplices)**.** *Let $Y = \Delta_{n_1} \times \cdots \times \Delta_{n_r}$ and let $f : Y \to \Delta_m$ be smooth. Then $T^k Y = \prod_j T^k \Delta_{n_j}$, and $T^2 f$ maps the fibre of $T^2 Y$ over x into $T^2 \Delta_m$ iff, for every output cell y with $f_y(x) = 0$, $D^2 f_y(x) \geq 0$ on $K_y \times K_y$, where $K_y = C_x \cap \ker Df_y(x)$ and $C_x = \prod_j C_{x_j}$.*

Proof. T preserves finite products and the lexicographic lemma holds row by row for any polyhedron [3, Lemma "the lexicographic lemma"]. The proof of [3, Theorem "second-order cone morphisms"] then runs block by block: $Df_y \geq 0$ on C_x, so in each block the gradient is constant on the support and no smaller on the empty cells, and the remaining steps are unchanged. □

Our implementation of lemma 14 rejects the counterexample above (TL5, negative control), so the passes reported below are not vacuous.

9 Tangent and provenance

Proposition 15 (T commutes with W)**.** *Let L be finite and discrete.*

(1) $T^k(L \times \Delta_X) = L \times T^k \Delta_X$: a source label has no tangent directions.

(2) $T^k \lambda : (l, c) \mapsto \delta_l \otimes c$, applied coefficient by coefficient, maps $L \times T^k \Delta_X$ into $T^k \Delta_{L \times X}$, and the unit, counit and comultiplication axioms of proposition 3 hold after applying T^k, for every k.

(3) Forgetting provenance, $\mathcal{D}\epsilon$, is a stochastic matrix, so it is a morphism at every order. At first order it maps C_q onto $C_{\mathcal{D}\epsilon(q)}$, with the lift of proposition 17(2) below.

Proof. (1) L is a finite disjoint union of points. (2) Each map in the three axioms is affine in the law, and T^k of an affine map is that map applied to every coefficient. The image is in $T^k \Delta_{L \times X}$ because the cells off row l have all coefficients 0, and the cells of row l carry the coefficients of c. (3) $\mathcal{D}\epsilon$ is marginalisation over L (proposition 13(1)); apply proposition 17. TL1 and TL3 check this. □

The multiplication axiom (2) of proposition 3 involves $\mathcal{D}\mathcal{D}X$, which is not a polyhedron, so the cone structure does not apply to it directly. Its restriction to n fixed sources is the pooling map of section 12.

Proposition 16 (the source face)**.** *Let $q = \lambda(l, p) = \delta_l \otimes p$. The map $\lambda(l, \cdot)$ embeds Δ_X as the face of $\Delta_{L \times X}$ of laws supported on $\{l\} \times X$.*

(1) $D\lambda_l : C_p \to C_q$ is injective. Its image is the set of $u \in C_q$ that vanish off row l: the directions along the face.

(2) *Every $u \in C_q$ decomposes uniquely as*

$$u = D\lambda_l(v) + m\,(Q' - q), \qquad v \in C_p, \quad m \geq 0, \quad Q' \in \Delta_{(L\setminus\{l\})\times X} \ (\text{if } m > 0),$$

where m is the mass u puts off row l. A feasible direction leaves the face exactly when it injects whole records under other labels.

(3) *After forgetting provenance, injection and drift are the same. The injection of records from Q under a new label l', the direction $\delta_{l'} \otimes Q - q$, and the drift of source l towards Q, the direction $\delta_l \otimes (Q - p)$, both map to $Q - p$ under $D\epsilon$. The relabelling direction $\delta_{l'} \otimes p - q$ is feasible and maps to 0.*

Proof. (1) $\lambda(l, \cdot)$ is affine and injective; its derivative puts v on row l. (2) q vanishes off row l, so $u \geq 0$ there. Let m be the sum of those entries and Q' their normalisation. Set $v = u_l + m\,p$, with u_l the row-l part of u. Then $\sum v = \sum u_l + m = 0$, and $v \geq 0$ wherever $p = 0$, because $u_l \geq 0$ there. The off-row part of u fixes m and Q', and then v, so the decomposition is unique. (3) $D\epsilon$ sums over labels: $\delta_{l'} \otimes Q - \delta_l \otimes p \mapsto Q - p$, $\delta_l \otimes (Q - p) \mapsto Q - p$, and $\delta_{l'} \otimes p - \delta_l \otimes p \mapsto 0$. TL2 checks (1)–(2) on 200 directions and TL3 checks (3). □

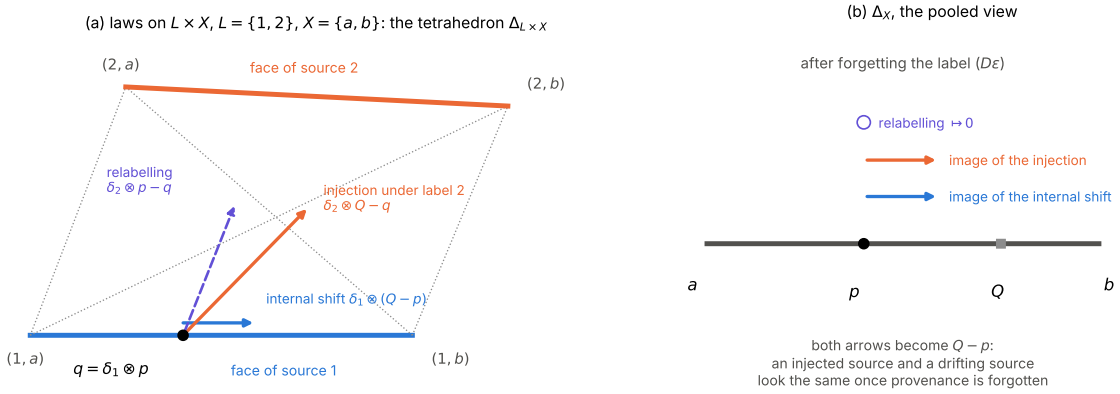


Figure 5: Proposition 16 with two sources and two outcomes. (a) The labelled laws form a tetrahedron; each source's laws form an edge (a face). At $q = \delta_1 \otimes p$ the internal shift stays on the face of source 1, while injection under label 2 and relabelling leave it. (b) After forgetting the label, the injection and the internal shift have the same image $Q - p$, and relabelling disappears.

So the provenance comonad separates two things that the pooled tangent space identifies: a source that drifts and a source that is added (fig. 5). This is the infinitesimal form of proposition 13(2). A screen on the pooled law alone can detect that data moved along $Q - p$, but not whether an existing source changed or a new one was injected. Only labelled tables tell these apart.

10 Tangent and marginalisation

Proposition 17 (marginalisation on feasible directions)**.** *Let S be a set of coordinates and M_S the marginalisation.*

(1) *M_S is a stochastic matrix, so $T^k M_S = M_S$ coefficient by coefficient maps $T^k \Delta_X$ into $T^k \Delta_{X_S}$ for every k, and $D^2 M_S = 0$.*

(2) *At first order it is* onto*: every* $v \in C_{M_S p}$ *is* $M_S u$ *for*

$$u(x) = v(x_S)\, p(x \mid x_S) \quad \text{if } M_S p(x_S) > 0, \qquad u(x) = v(x_S)/|\pi_S^{-1}(x_S)| \quad \text{otherwise,}$$

and this u *lies in* C_p*.*

(3) *Its kernel on directions,* $\{u : \sum u = 0,\ M_S u = 0\}$*, has dimension* $|X| - |X_S|$*. These are the directions the window* X_S *cannot see; for a cover they are the common kernel of all its windows, the hidden channels of [1, Theorem F].*

Proof. (1) [3, Proposition "morphisms"]. (2) $M_S u(x_S) = v(x_S) \sum_{x \mapsto x_S} p(x \mid x_S) = v(x_S)$ in the first case, and the same in the second. Feasibility: if $p(x) = 0$ and $M_S p(x_S) > 0$ then $u(x) = 0$; if $M_S p(x_S) = 0$ then $v(x_S) \geq 0$, so $u(x) \geq 0$. (3) M_S is onto the zero-sum directions on X_S, so the kernel dimension is $(|X| - 1) - (|X_S| - 1)$. TL4 checks all three on 200 laws with zeros. □

Example 18 (a direction the window cannot see)**.** In example 12, order the cells $(R, H) = (1,1), (1,0), (0,1), (0,0)$ and take $u = (1, -1, 0, 0)$: move records from "recovered, low risk" to "recovered, high risk". The R window sees nothing: $M_{\{R\}} u = 0$. But the answer of example 12 moves fast. By proposition 8(5), with $p(R{=}1) = \frac{23}{40}$, the derivative of $P(H{=}1 \mid R{=}1)$ along u is $u(1,1)/p(R{=}1) = \frac{40}{23}$. A window that misses a coordinate misses exactly the directions to which a conditional on that coordinate responds.

11 Tangent and conditioning

Proposition 19 (conditioning on feasible directions)**.** *Let* $p(A) > 0$*.*

(1) $DC_A(p)$ *maps* C_p onto $C_{C_A p}$*. A lift of* $v \in C_{C_A p}$ *is* $u = p(A)\, v$ *(zero off* A*).*

(2) *The kernel of* $DC_A(p)$ *on directions is* $\{u : \sum u = 0,\ N_A u = u(A)\, C_A p\}$*, of dimension* $|X| - |A|$*. These directions change how much mass* A *carries, or move mass outside* A*, without changing its shape inside* A*.*

(3) C_A *passes the second-order corner criterion at every point of its domain, corners included [3, Corollary "the second-order subcategory"].*

(4) *Near the wall the lift shrinks: the move of* p *needed for a given move* v *of* $C_A p$ *has size* $p(A)\,|v|$*, so* $\|DC_A(p)\| \sim 1/p(A)$*. At the wall itself* C_A *is undefined and has no continuous extension (proposition 10).*

Proof. (1) With $DC_A(p)u = (N_A u - u(A) C_A p)/p(A)$ (proposition 8(5)): if $p(x) = 0$ with $x \in A$, then $C_A p(x) = 0$ and $u(x) \geq 0$, so the image is ≥ 0 there. The image sums to 0. For the lift, $u = p(A)v$ has $u(A) = 0$, so $DC_A(p)u = v$; it is feasible because the zeros of $C_A p$ in A are zeros of p. (2) Read off from the formula; the condition fixes $N_A u$ up to the scalar $u(A)$, leaving $1 + (|X| - |A|)$ free coordinates minus the sum constraint. (4) From (1) and proposition 8(5). TL5 checks (1), (2) and (4), and re-checks (3) with lemma 14 at three corners and with 900 sampled points of $T^2\Delta$. □

In the worked example, moving the high-risk share among the recovered by 0.01 takes a move of only $0.01 \cdot \frac{23}{40} \approx 0.006$ in the joint law, made inside $R = 1$. Were recovery rare, say $p(R{=}1) = 0.01$, the same move would take 0.0001. Conditioning on a rare event amplifies small injections, and proposition 19(4) is the exact rate.

12 Tangent and pooling

Pooling n sources is the restriction of the monad multiplication to n fixed sources:

$$\mu_n : \Delta_n \times (\Delta_X)^n \to \Delta_X, \qquad \mu_n(w, p_1, \dots, p_n) = \sum_i w_i p_i = \mathcal{D}\epsilon\Big(\sum_i w_i\, \lambda(i, p_i)\Big).$$

It is bilinear, not affine, so the fact that stochastic matrices are morphisms does not cover it. The next theorem does, at every order. Write $H = [0, \infty)$ for the half-line.

Theorem 20 ($T^k H$ is generated by elementary tensors)**.** *For every k, $T^k H \subset W_k$ is the convex cone generated by the products $u_1(e_1)\, u_2(e_2) \cdots u_k(e_k)$ with each $u_i = x_i + y_i e_i \in TH$ (that is, $x_i > 0$, or $x_i = 0 \le y_i$). Consequently $T^k H$ is closed under the product of W_k.*

Proof. Write G_k for the cone generated by the elementary tensors. Induction on k; $G_0 = T^0 H = H$.

$G_k \subseteq T^k H$: a point of $T^k H = T(T^{k-1}H)$ is $\alpha + \beta e_k$ with $\alpha + s\beta \in T^{k-1}H$ for all small $s \ge 0$. For $g \in G_{k-1}$ and $u_k = x + y e_k$: if $x > 0$ then $g\, x + s\, g\, y = (x + sy)g \in T^{k-1}H$ for small s; if $x = 0 \le y$ then $0 + s\, yg \in T^{k-1}H$. $T^k H$ is a convex cone.

$T^k H \subseteq G_k$: let $\alpha + \beta e_k \in T^k H$ and fix $s > 0$ with $\alpha + s\beta \in T^{k-1}H$. Then

$$\alpha + \beta e_k = \alpha\left(1 - \tfrac{1}{s} e_k\right) + \left(\beta + \tfrac{1}{s}\alpha\right) e_k.$$

Here $\alpha \in T^{k-1}H$, and $\beta + \frac{1}{s}\alpha = \frac{1}{s}(\alpha + s\beta) \in T^{k-1}H$ because it is a cone; both lie in G_{k-1} by induction. $1 - \frac{1}{s} e_k$ and e_k lie in TH. So the right side is in G_k.

Products: $W_k = W_1^{\otimes k}$ as algebras, so the product of two elementary tensors is $\prod_i (u_i v_i)(e_i)$, and TH is closed under products: $(x + ye)(x' + y'e) = xx' + (xy' + x'y)e$, which lies in TH when both factors do. By bilinearity the cone G_k is closed under products. □

Corollary 21 (pooling is a morphism at every order)**.** *(1) The product of laws $(x, y) \mapsto x \otimes y$ and the labelled joint $J(w, p_1, \dots, p_n) = \sum_i w_i\, \lambda(i, p_i)$ map T^k of their domains into T^k of their codomains, for every k.*

(2) So does pooling, $\mu_n = \mathcal{D}\epsilon \circ J$. Its derivative maps the product of feasible cones into C_{μ_n}, including at sources of weight 0 and cells where every source vanishes. $D^3 \mu_n = 0$.

Proof. (1) Each cell of $x \otimes y$ is a product $x_i y_j$, and each cell (i, x) of J is $w_i p_i(x)$. By theorem 20 every cell of the image is in $T^k H$. The equations hold because $\sum_x p_i(x) = 1$ and $\sum_i w_i = 1$ hold in W_k. (2) $\mathcal{D}\epsilon$ is stochastic; μ_n is bilinear, so its third derivative vanishes. □

Corollary 21(1) also proves the statement of [3, Proposition "morphisms"] about products of laws, which was checked there but not proved. TL9 checks theorem 20 directly: an explicit decomposition into elementary tensors for 1510 points of $T^k H$, $k \le 4$, and closure under products for all pairs of a grid when $k = 2$ and 100,000 sampled pairs when $k = 3$. TL6 checks the second-order criterion for μ_n at corners (lemma 14), including a source of weight 0. At the level of individual cells the reason is simple. If $w_i > 0$ at a cell where the pool vanishes, then p_i vanishes there and the first-order term $w_i u_i$ keeps it non-negative. If $w_i = 0$, the term $a_i p_i$ does. If both vanish, every entry is non-negative.

Remark 22 (where the pooling paradox lives)**.** The tangent companion shows that two data sets choosing the same act can pool to one choosing the other, through the curvature of the decision boundary [3, Proposition "pooling paradox = curvature"]. Corollary 21 locates the paradox: pooling itself is as tame as a Markov kernel at every order. Of the maps met so far, only conditioning (partial, with a pole) and the decision functional bend.

13 Differentiating the contextual Bayes decomposition

Theorem 11 expresses $C_A \circ \mu_n$ through the Bayes weights. Differentiating it splits every sensitivity of a pooled conditional into two terms.

Proposition 23 (reweighting and within-source terms)**.** *Let $\sum_j \pi_j p_j(A) > 0$, and move the weights along a and the sources along $u_1, \dots, u_n$. Then*

$$D(C_A \circ \mu_n)[a,u] = \underbrace{\sum_l \dot{\pi}_l^A \, C_A(p_l)}_{\textit{reweighting}} + \underbrace{\sum_l \pi_l^A \, DC_A(p_l)[u_l]}_{\textit{within sources}},$$

$$\dot{\pi}_l^A = \frac{a_l p_l(A) + \pi_l u_l(A)}{Z} - \pi_l^A \, \frac{\sum_j \big(a_j p_j(A) + \pi_j u_j(A)\big)}{Z},$$

with $Z = \sum_j \pi_j p_j(A)$ (sources with $p_l(A) = 0$ drop out of the second sum). If only the weights move, only the reweighting term remains; if only the sources move and $p_l(A)$ is the same for all l and stays so, only the within-source term remains.

Proof. Differentiate $C_A(\sum \pi_l p_l) = \sum_l \pi_l^A C_A(p_l)$ (theorem 11) by the product rule; $\dot{\pi}^A$ is the derivative of $\pi_l p_l(A)/Z$. TL7 checks the identity on 100 random configurations. □

Example 24 (how fast the answer moves with the small-trial share)**.** In example 12, let the small-trial share π_1 vary, with $\pi_2 = 1 - \pi_1$. The answer is

$$f(\pi_1) = P(H{=}1 \mid R{=}1) = \frac{\frac{1}{5} + \frac{1}{10}\pi_1}{\frac{1}{2} + \frac{3}{10}\pi_1},$$

$$f(\tfrac{1}{4}) = \tfrac{9}{23}, \qquad f'(\tfrac{1}{4}) = -\tfrac{16}{529} \approx -0.030, \qquad f''(\tfrac{1}{4}) = \tfrac{384}{12167} \approx 0.032.$$

Only the weights move, so proposition 23 gives $f' = \dot{\pi}_1^A \, (\frac{3}{8} - \frac{2}{5})$ with $\dot{\pi}_1^A = \frac{640}{529} \approx 1.21$: the Bayes weight moves 1.21 times as fast as the prior weight, and the answer moves by that times the gap between the sources' answers. Raising the small-trial share from 25% to 26% lowers the answer by about 0.0003, and the second-order term adds about 0.0000016. Since f is a ratio of affine functions, the answer stays between the two sources' answers, $\frac{3}{8}$ and $\frac{2}{5}$, for every share (fig. 6; TL8, exact).

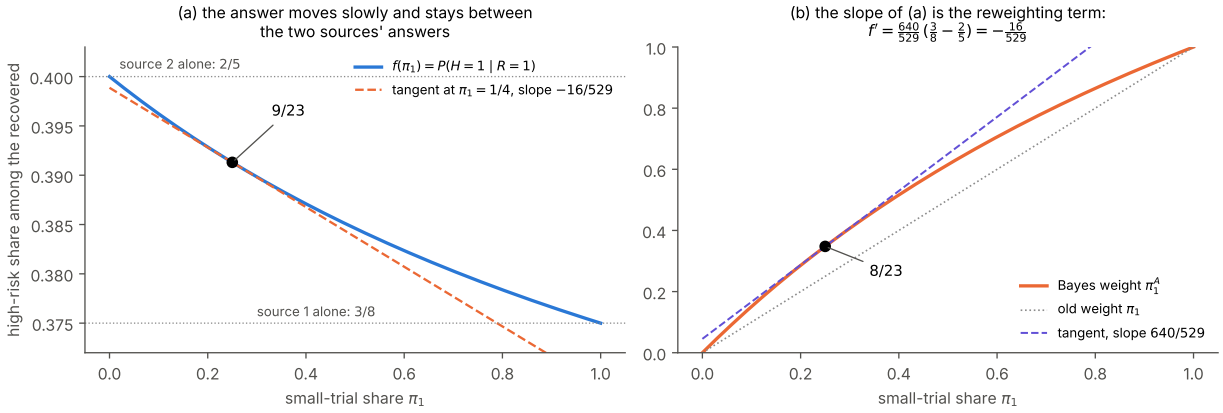


Figure 6: Example 24. (a) The answer $P(H{=}1 \mid R{=}1)$ as the small-trial share varies, with its tangent at $\frac{1}{4}$. (b) The Bayes weight of the small trials, against the old weight. The slope of (a) is the slope of (b) times the gap $\frac{3}{8} - \frac{2}{5}$ between the sources' answers.

14 Strata and jumps

Derivatives describe gradual change. Conditioned probabilities can also change suddenly, and a source can drop out of the explanation of an event. This section says exactly when this happens, for finite sets. It has three results.

- The sources that survive conditioning are explicit, and they cut the space of labelled laws into strata (theorem 25).
- Inside the domain of conditioning nothing jumps. The rate of change is at most $2/P(A)$, so a sudden change needs the event to become rare (theorem 27(1)–(2)).
- At the wall $P(A) = 0$ the limit depends on the leading jet of the path, not on the point reached. A smooth (analytic) passage through the wall is removable; a jump needs a kink at the wall, a non-analytic path, or a jump of the law itself (theorem 27(3)–(5)).

The last point is where points give way to functions. At the wall, the conditioned state is a function of how the path arrives, its germ, and not of where it arrives.

The state and the operator. A source-labelled state $\Theta = (\pi_l, p_l)_{l \in L}$ is the labelled law $\Lambda = \sum_l \pi_l \, \lambda(l, p_l) \in \Delta_{L \times X}$, with pooled law $P = \mathcal{D}\epsilon(\Lambda)$. Conditioning the state is

$$\mathfrak{C}_A = C_{L \times A} : \Delta_{L \times X} \rightharpoonup \Delta_{L \times X}.$$

By theorem 11, the L-marginal of $\mathfrak{C}_A(\Lambda)$ is the vector of Bayes weights π^A, and its row l, normalised, is $C_A(p_l)$.

Theorem 25 (provenance support)**.** *Let $P(A) = \sum_l \pi_l p_l(A) > 0$.*

(1) The surviving sources are $\operatorname{supp} \pi^A = L_A(\Lambda) = \{l : \pi_l p_l(A) > 0\}$*. A source drops out if it has weight* 0*, or if it gives A probability* 0*.*

(2) $\pi_l^A = \pi_l p_l(A)/P(A)$*, and for surviving l, j the posterior odds are the prior odds times the likelihood ratio:* $\pi_l^A / \pi_j^A = (\pi_l/\pi_j)\,(p_l(A)/p_j(A))$*.*

(3) Forgetting commutes with conditioning: $\mathcal{D}\epsilon \circ \mathfrak{C}_A = C_A \circ \mathcal{D}\epsilon$, *so* $\sum_l \pi_l^A C_A(p_l) = C_A(P)$.

(4) A source with $\pi_l > 0$ *and* $p_l(A) = 0$ *has no row in* $\mathfrak{C}_A(\Lambda)$. *Its own conditional* $C_A(p_l)$ *is undefined, and nothing requires it.*

So the domain of $\mathfrak{C}_A$ *is the union of the strata* $\mathcal{S}_J(A) = \{\Lambda : L_A(\Lambda) = J\}$ *for non-empty* $J \subseteq L$. *On each stratum* $\mathfrak{C}_A$ *is a rational map with a positive denominator, and the wall is* $\mathcal{S}_\varnothing(A)$.

Proof. $\mathfrak{C}_A(\Lambda)(l, x) = \pi_l p_l(x)\mathbf{1}_A(x)/P(A)$. Summing over x gives (1) and (2), and the odds follow. Summing over l gives (3). Row l vanishes iff $\pi_l p_l(A) = 0$, which is (4). SJ1 checks all four exactly on 291 random families with zero weights and zero likelihoods. □

Proposition 26 (four questions)**.** *Let* $P(A) > 0$ *and* $n = |L|$.

(1) When are the weights preserved? $\pi^A = \pi$ *iff* $p_l(A) = P(A)$ *for every* l *with* $\pi_l > 0$.

(2) When do conditioning and pooling commute? *Suppose every source of positive weight has* $p_l(A) > 0$. *Pooling the conditioned sources with the old weights gives* $C_A(P)$ *iff* $\sum_l (\pi_l^A - \pi_l)\, C_A(p_l) = 0$. *This follows from (1). It is equivalent to (1) when the* $C_A(p_l)$ *are affinely independent. It holds whatever the weights do when the* $C_A(p_l)$ *coincide.*

(3) What happens to the fibres of forgetting? $\mathfrak{C}_A$ *maps the fibre* $\mathcal{D}\epsilon^{-1}(P)$ *onto the fibre* $\mathcal{D}\epsilon^{-1}(C_A P)$*: a labelled law* Λ' *over* $C_A P$ *comes from* $\Lambda = P(A)\Lambda' + \delta_{l_0} \otimes P|_{X \setminus A}$. *The dimensions split as*

$$\underbrace{(n-1)\,|\operatorname{supp} P|}_{\text{fibre over } P} = \underbrace{(n-1)\,|\operatorname{supp} P \cap A|}_{\text{fibre over } C_A P} + \underbrace{(n-1)\,|\operatorname{supp} P \setminus A|}_{\text{lost by conditioning}}.$$

Conditioning keeps the provenance ambiguity on A *and discards the provenance off* A. *It never resolves ambiguity.*

(4) Can provenance be recovered after conditioning? *Not from the pooled law: every* $w \in \Delta_L$ *is the posterior provenance of some state in the fibre over* P *(lift* $\sum_l w_l\, \lambda(l, C_A P)$ *by (3)). With known source laws and unknown weights,* π^A *is determined by* $C_A(P)$ *iff the* $C_A(p_l)$ *with* $p_l(A) > 0$ *are linearly independent. If they are, so are those* p_l*: conditioning can destroy identifiability but cannot create it.*

Proof. (1) By theorem 25(2), $\pi_l^A = \pi_l$ iff $p_l(A) = P(A)$ or $\pi_l = 0$. (2) The old pool minus $C_A(P)$ is $\sum_l (\pi_l - \pi_l^A) C_A(p_l)$ by theorem 25(3). The weight differences sum to 0, so this is an affine relation among the $C_A(p_l)$. (3) $\mathcal{D}\epsilon(\Lambda) = P(A)\, C_A P + P|_{X \setminus A} = P$, and on $L \times A$ the law Λ is $P(A)\Lambda'$, so $\mathfrak{C}_A(\Lambda) = \Lambda'$. A fibre of forgetting over Q is a product, over $x \in \operatorname{supp} Q$, of simplices of dimension $n - 1$ (how $Q(x)$ is split among the sources). The preimages of a fixed Λ' are free exactly in the cells off A. (4) The first claim is (3). Mixture weights are determined by the mixture iff the components are linearly independent, which for laws is the same as affinely independent, because a linear relation among laws has coefficients summing to 0. If $\sum_l c_l p_l = 0$, then restricting to A gives $\sum_l c_l\, p_l(A)\, C_A(p_l) = 0$, so independence of the $C_A(p_l)$ forces every $c_l p_l(A) = 0$, hence $c_l = 0$. SJ2 and SJ3 check all four, (3) with explicit lifts and with ranks of sampled fibres. □

For example, take $p_1 = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, 0)$ and $p_2 = (\frac{3}{8}, \frac{3}{8}, 0, \frac{1}{4})$ on $\{a, b, c, d\}$ with $A = \{a, b\}$, and equal weights. Conditioning moves the weights to $(\frac{2}{5}, \frac{3}{5})$, yet pooling with the old weights is still correct, because both sources condition to the same law. The two sources are distinguishable before conditioning and indistinguishable after it.

Theorem 27 (regimes of conditioned probability)**.** *Let $t \mapsto \Lambda_t$ be a path in $\Delta_{L\times X}$ and write $P_t(A) = \Lambda_t(L \times A)$.*

(1) Regular regime. $\|D\mathfrak{C}_A(\Lambda)h\|_1 \le 2\|h\|_1/\Lambda(L\times A)$. *So if Λ_t is continuous (or C^k) and $P_t(A) \ge \varepsilon > 0$, then $\mathfrak{C}_A(\Lambda_t)$, $\pi^A(t)$ and $C_A(P_t)$ are continuous (or C^k), and*

$$\|\pi^A(t_1) - \pi^A(t_0)\|_1 \le 2\int_{t_0}^{t_1} \frac{\|\dot\Lambda_t\|_1}{P_t(A)}\,dt.$$

A complete change of the source that explains A ($\|\Delta\pi^A\|_1 = 2$) needs $\int \|\dot\Lambda_t\|_1/P_t(A)\,dt \ge 1$. At speed at most v, with $P_t(A) \ge \varepsilon$, that takes time at least ε/v.

(2) Changes of stratum. *If $P_t(A)$ stays positive while a source leaves or joins L_A, then $\pi^A_l(t)$ tends to 0 or leaves it continuously. The leaving source's own conditional $C_A(p_l(t))$ may have no limit, because that source is at its own wall, but it carries vanishing weight. The conditioned state can be continuous while a source's own conditional jumps.*

(3) The wall, one side. *Let $\Lambda_{t^*+s} = \sum_j c_j s^j$ for small $s \ge 0$ (a polynomial or convergent series), with $P_{t^*}(A) = 0$ and $P_t(A) \not\equiv 0$. Let m be the least j with $c_j|_{L\times A} \ne 0$. Then $c_m|_{L\times A} \ge 0$, and*

$$\lim_{s\downarrow 0} \mathfrak{C}_A(\Lambda_{t^*+s}) = C_{L\times A}(c_m),$$

with error $O(s)$. The limiting provenance is the Bayes weighting of the leading jet. The sources that survive are those whose mass on A vanishes at the least order m; the others become extinct at the wall. Different jets give different limits, and every law on $L \times A$ occurs (proposition 10).

(4) Analytic passages are removable. *If Λ_t is analytic on both sides of t^* and $P_t(A) \not\equiv 0$, then $\mathfrak{C}_A(\Lambda_t)$ extends analytically through t^*, and its two one-sided limits agree.*

(5) Where jumps come from. *So if $\mathfrak{C}_A(\Lambda_t)$ jumps at t^*, then one of the following holds:*

- *the law Λ_t itself jumps;*
- *the path reaches the wall with a kink, and the conditionals of its two one-sided leading jets differ;*
- *the path is not analytic at the wall, and then the limit need not exist.*

Proof. (1) On $L \times X$, $D\mathfrak{C}_A(\Lambda)h = (N_A h - h(L\times A)\,\mathfrak{C}_A\Lambda)/\Lambda(L\times A)$ (proposition 8(5)), and $\|N_A h\|_1 + |h(L\times A)| \le 2\|h\|_1$. Taking the L-marginal, a stochastic map, does not increase $\|\cdot\|_1$. Then integrate. (2) $\pi^A_l = \Lambda_t(l, A)/P_t(A)$ is continuous, and so is the rest, by (1). (3) Each cell of $L\times A$ is non-negative, so its first non-zero coefficient is positive. Below order m every such coefficient is 0, so $\mathfrak{C}_A(\Lambda_{t^*+s}) = (s^m c_m|_{L\times A} + O(s^{m+1}))/(s^m c_m(L\times A) + O(s^{m+1}))$. (4) A non-negative analytic function vanishing at t^* is $(t-t^*)^{m_c}h_c(t)$ with m_c even and $h_c(t^*) > 0$. Divide numerator and denominator by $(t-t^*)^m$, $m = \min m_c$. The denominator becomes $\sum_{m_c=m} h_c(t^*) > 0$ at t^*. (5) combines (1)–(4).

SJ4 checks (1) on 100 random paths (the largest ratio to the bound is 0.42) and (2) on an oscillating extinction. SJ5 checks (3) exactly on 200 polynomial paths, (4) on a two-sided passage, and the kink of (5). □

Example 28 (gradual, then sudden)**.** Two sources of weight $\frac{1}{2}$ each. Source 1 produces a or c, source 2 produces b or c, and $A = \{a, b\}$ ("a flagged record"). Over time, source 1 stops producing flagged records and source 2 starts:

$$\Lambda_t(1, a) = \eta + \max(-t, 0), \qquad \Lambda_t(2, b) = \eta + \max(t, 0), \qquad t \in [-\tfrac{1}{4}, \tfrac{1}{4}],$$

with the rest of each source's mass on c. The source weights never change, and $P_t(A) = 2\eta + |t|$ changes gradually (fig. 7(a)). But the source that explains a flagged record switches:

$$\pi_1^A(t) = \frac{\eta + \max(-t, 0)}{2\eta + |t|}$$

falls from $\frac{3}{4}$ to $\frac{1}{4}$ over an interval of length exactly 4η, with slope $-1/(4\eta)$ at $t = 0$ (fig. 7(b)). The smaller the gap η between the path and the wall, the more sudden the switch, as theorem 27(1) requires. At $\eta = 0$ the path touches the wall with a kink: the one-sided jets are $\delta_{(1,a)}$ and $\delta_{(2,b)}$, and π_1^A jumps from 1 to 0, a complete replacement of the explaining source. A smooth path into the wall behaves differently. Along $p_t = (t,\ t(1-t),\ 1-t-t(1-t))$ on $\{a, b, c\}$, the event $\{a, b\}$ becomes impossible at $t = 0$. Yet $C_A(p_t)$ tends to $(\frac{1}{2}, \frac{1}{2}, 0)$, the conditional of the leading jet $(1, 1, 0)$. Here $C_A(p_t) = (\frac{1}{2-t}, \frac{1-t}{2-t}, 0)$ moves at a bounded rate all the way in (SJ5, SJ6).

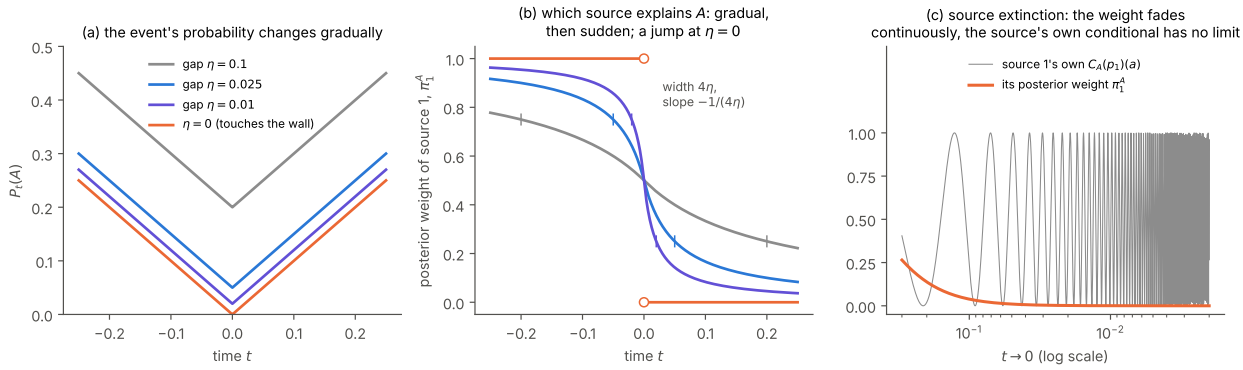


Figure 7: Example 28 and theorem 27. (a) $P_t(A)$ changes gradually for every gap η. (b) The posterior weight of source 1 switches over a window of width 4η (ticks at $\pm 2\eta$), and jumps at $\eta = 0$, where the path meets the wall with a kink. (c) Source extinction inside the domain. Source 1's posterior weight fades continuously to 0, while its own conditional oscillates without a limit (SJ4).

regime	the law	$P(A)$	the conditioned state	checks
regular	continuous	$\geq \varepsilon$	continuous, rate $\leq 2/\varepsilon$	SJ4
change of stratum	continuous	$\geq \varepsilon$	continuous; a leaving source's own conditional may jump	SJ4
near the wall	continuous	small, > 0	continuous but steep: gradual, then sudden	SJ6
wall, analytic	analytic	$\to 0$	removable; limit = conditional of the leading jet	SJ5
wall, kink	one-sided analytic	$\to 0$	jumps iff the one-sided jets condition differently	SJ5, SJ6
outside jump	jumps	any	may jump	—

Remark 29 (transition operators and what is not covered)**.** An outside jump replaces Λ by $J(\Lambda)$ for a map J of $\Delta_{L \times X}$. If J is a stochastic matrix it preserves total probability and positivity. It is *provenance-consistent*, preserving every source weight π_l, when it acts row by row through a source-indexed kernel: a morphism of the two-sided Kleisli category (proposition 4). Even then

π^A can jump, because the $p_l(A)$ can. The hybrid dynamics of regular flows interrupted by such operators is not developed here. Neither is the analogue of theorem 27 beyond finite sets, where the answer depends on the notion of conditional probability and on the topology, and needs separate care.

15 Step 2 in one table

operation	domain	first order on cones	order 2	every order
$\mathcal{D}\epsilon$, M_S, $\lambda(l,\cdot)$	total	onto (explicit lift)	yes	yes (stochastic)
μ_n (pooling)	total	into	yes	yes (corollary 21)
C_A (conditioning)	$p(A) > 0$	onto, pole $1/p(A)$	yes	yes (theorem 57, Step 4e)
$\mathfrak{C}_A$ (labelled)	$P(A) > 0$, strata $\mathcal{S}_J$	continuous, rate $\leq 2/P(A)$	yes	yes on the domain; at the wall: the leading jet (theorem 27)

Conditioning beyond order two was left open when Step 2 was written. The tangent companion gives only a necessary condition at order k [3, Proposition "order k: necessity, and order three"]. Step 4e settles it: conditioning is subtraction-free, so it keeps feasibility at every order (theorem 57).

16 Step 3: the order of a transition

Theorem 27(3) shows that at the wall the conditioned state is decided by the leading term of the path, not by the point it reaches. So a transition carries more structure than "continuous or not". Each source approaches the event at its own *order*, and the orders decide which sources survive, how fast the others disappear, whether the two sides of a transition agree, and how sudden a near-miss is. This step makes that precise. The main tool is one observation: on non-negative quantities, "order" turns sums into minima and products into sums. Positivity is what makes this exact.

Germs and orders. We look at quantities near a time t^* from one side and write $s = t - t^* \downarrow 0$. A *germ* is a function of s for small $s > 0$, up to equality near 0. To compare germs we need them to be comparable, and the natural setting is a *Hardy field* $\mathcal{H}$: a field of germs of differentiable functions closed under differentiation [13, 14]. Every element of a Hardy field has a limit in $[-\infty, \infty]$, so the ratio of two non-zero elements always has a limit. Examples are the generalised power sums $\sum_i c_i s^{a_i}$, Hardy's exp-log functions (such as $s^{3/2}$, $e^{-1/s}$, $1/\log\frac{1}{s}$), and the germs definable in an o-minimal structure. Write $\mathcal{H}_+$ for the positive germs together with 0. For $f, g \in \mathcal{H}_+ \setminus \{0\}$ write $f \asymp g$ if f/g tends to a finite non-zero limit. The *order* $v(f)$ is the class of f, ordered by $v(f) < v(g)$ iff $g/f \to 0$ (so g vanishes faster); set $v(0) = \infty$. For $f \sim c\, s^a$ with $c > 0$, $v(f)$ is the exponent a and c is the *leading coefficient*. Flat germs such as $e^{-1/s}$ have orders beyond every power: $v(s^n) < v(e^{-1/s})$ for all n.

Theorem 30 (orders tropicalise)**.** *On $\mathcal{H}_+$,*

$$v(f+g) = \min\{v(f), v(g)\}, \qquad v(fg) = v(f) + v(g).$$

If $f \sim c\, s^a$ and $g \sim d\, s^b$, then $fg \sim cd\, s^{a+b}$, and $f + g \sim c\, s^a$ if $a < b$, $d\, s^b$ if $b < a$, and $(c+d)\, s^a$ if $a = b$. So v is a homomorphism from the semiring $(\mathcal{H}_+, +, \times)$ onto the tropical semiring $(\Gamma \cup \{\infty\}, \min, +)$ of orders. Positivity is necessary: $s + s^2$ and $-s$ have order 1, and their sum has order 2.

Proof. For positive f, g, $\max(f,g) \leq f + g \leq 2\max(f,g)$, so $f + g \asymp \max(f,g)$, the germ of least order. On a tie the leading coefficients are both positive and add without cancelling. Products

are immediate. OR1 checks this exactly on 500 pairs of non-negative power sums, and checks the counterexample. □

Corollary 31 (the order calculus of the layers)**.** *Let a labelled path $\Lambda_s(l,x) = \pi_l(s)\, p_{l,s}(x)$ have all its cells in $\mathcal{H}_+$. Write*

$$m_l = v\big(\Lambda_s(l,A)\big) = v(\pi_l) + v\big(p_{l,s}(A)\big), \qquad m = v\big(P_s(A)\big) = \min_l m_l.$$

operation	*order*	*leading coefficient*
labelling $\pi_l p_l(x)$	$v(\pi_l) + v(p_l(x))$	*product*
pooling $P(x) = \sum_l \Lambda(l,x)$	$\min_l v(\Lambda(l,x))$	*sum over the sources of least order*
marginalisation $M_S P(x_S)$	*minimum over the fibre*	*sum over the cells of least order*
conditioning $\mathfrak{C}_A \Lambda(l,x)$	$v(\Lambda(l,x)) - m$	*divided by the leading coefficient of* $P(A)$
Bayes weight π_l^A	$m_l - m$	c_l/c

Here c_l and c are the leading coefficients of $\Lambda_s(l,A)$ and $P_s(A)$.

Proof. Every operation is built from sums and products of non-negative germs, with one division by $P_s(A)$. Apply theorem 30. OR2 checks every row exactly on 200 random labelled paths. □

Theorem 32 (the hierarchy of transitions)**.** *Let the cells of Λ_s on $L \times A$ lie in one Hardy field, not all zero. Then $\mathfrak{C}_A(\Lambda_s)$ has a limit as $s \downarrow 0$. Its support is the set of cells of least order m, and the surviving sources are $\{l : m_l = m\}$. When the cells are asymptotic to powers, the limit is the normalised vector of leading coefficients of order m. In particular:*

(1) Common order. *Several sources reach the event at the least order. All survive, with limiting weights $c_l / \sum_{j:m_j=m} c_j$.*

(2) Unequal order. *A source with $m_l > m$ disappears from the explanation, at rate $\pi_l^A(s) \sim (c_l/c)\, s^{m_l - m}$. Its* extinction order *is $m_l - m$. For generalised power sums, the conditioned state converges at rate $O(s^\delta)$, where δ is the gap from m to the next exponent appearing on $L \times A$.*

(3) One-sided mismatch. *The two sides of t^* carry independent data $(m_l^\pm, c_l^\pm)$. The one-sided limits differ iff the two leading conditionals differ. If the path is analytic across t^*, every order on $L \times A$ is even and the two sides agree (theorem 27(4)). Infinite differentiability is not enough: with $\Lambda_t(1,a) = e^{-1/t^2}$ for $t < 0$ and $\Lambda_t(2,b) = e^{-1/t^2}$ for $t > 0$ (zero otherwise), every derivative exists, and the explaining source jumps from 1 to 2.*

(4) Flat approach. *Germs with every derivative zero at $s = 0$ still have orders, beyond all powers, and inside a Hardy field the limit exists. For example, $e^{-1/s}/\log\frac{1}{s}$ against $e^{-1/s}$ has limiting share 0, reached only logarithmically slowly. Outside a Hardy field the limit can fail to exist: $e^{-1/s}(1 + \sin\frac{1}{s})$ against $e^{-1/s}$ has a share that oscillates in $[0, \frac{2}{3}]$.*

So whether a conditioned probability has a limit is a question of comparability, and what the limit is a question of order.

Proof. In a Hardy field each ratio $f_c/f_{c'}$ of positive cells has a limit in $[0,\infty]$. Choose a cell c^* of least order. Then $f_c/f_{c^*} \to r_c < \infty$ for every c, with $r_c > 0$ iff $v(f_c) = m$. So $\mathfrak{C}_A(\Lambda_s)(c) = (f_c/f_{c^*}) / \sum_{c'}(f_{c'}/f_{c^*}) \to r_c / \sum r_{c'}$, and the denominator is at least 1. For powers, r_c is the ratio of leading coefficients. (1) and (2) are corollary 31. The rate in (2) comes from expanding numerator

and denominator one exponent further. (3) The two sides are separate germs; the analytic case is theorem 27(4); the flat example is C^∞ because e^{-1/t^2} is. (4) $e^{-1/s}$ and $\log \frac{1}{s}$ lie in Hardy's exp-log field. $1+\sin \frac{1}{s}$ is not eventually comparable with 1. Checks: OR3 for (1)–(2), exactly on 200 paths plus a fractional example; OR4 for (3); OR5 for (4), computed in log space. □

Proposition 33 (the order is the level of the tangent tower that sees the transition)**.** *Let* $\Lambda_s = \sum_j c_j s^j$ *be analytic with* $P_0(A) = 0$*, and let* $j^k\Lambda = \sum_{j\le k} c_j s^j$ *be its* k*-jet.*

(1) For $k < m$ *the* k*-jet has no mass on* $L \times A$*, so it cannot be conditioned. For* $k \ge m$ *it has the same limit as the path.*

(2) In the Weil-algebra picture of Step 2, the k*-jet is the point* $\Lambda(e_1 + \cdots + e_k)$ *of* $T^k\Delta_{L\times X}$*. Its coefficient on* e^S *is* $|S|!\, c_{|S|}$*. On* $L \times A$ *the first non-zero coefficients sit at* $|S| = m$*, which are the coefficients the lexicographic lemma reads first.*

So a transition of order m *is invisible to* T^k *for* $k < m$ *and determined at level* m *of the tower.*

Proof. (1) The coefficients of order $< m$ vanish on $L \times A$, and the limit depends only on c_m (theorem 32). (2) $(e_1 + \cdots + e_k)^j = j! \sum_{|S|=j} e^S$ because $e_i^2 = 0$. OR6 checks (1) on 200 paths for $k \le 5$, and (2) with the Weil-algebra arithmetic. □

Example 34 (order and suddenness)**.** Repeat example 28 at order m: $\Lambda_t(1, a) = \eta + \max(-t, 0)^m$ and $\Lambda_t(2, b) = \eta + \max(t, 0)^m$. Then $\pi_1^A(t) = (\eta + \max(-t, 0)^m)/(2\eta + |t|^m)$ runs from $\frac{3}{4}$ to $\frac{1}{4}$ over $[-(2\eta)^{1/m}, (2\eta)^{1/m}]$ exactly. For the same gap $\eta = \frac{1}{1000}$, the switch has width 0.004 at order 1, 0.089 at order 2, and 0.25 at order 3. A higher-order approach is more gradual. For $m \ge 2$ the curve is flat at the crossing, where both sources explain the event equally, and the change happens on either side of it (fig. 8(b); OR7, exact). At $\eta = 0$ every order gives the same jump from 1 to 0: the order shapes the approach, not the jump.

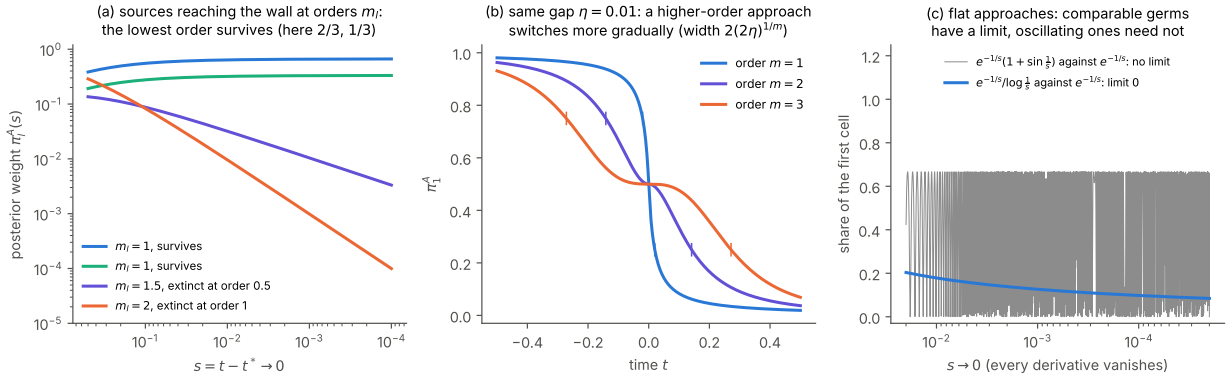


Figure 8: Orders of transitions. (a) Four sources reach the wall at orders $1, 1, \frac{3}{2}, 2$. The two of order 1 survive with weights $\frac{2}{3}, \frac{1}{3}$, and the others vanish at their extinction orders $\frac{1}{2}$ and 1, the slopes on log scales (theorem 32(1)–(2)). (b) Example 34: the same gap, approached at orders $1, 2, 3$ (ticks at $\pm(2\eta)^{1/m}$). (c) Flat approaches: a comparable pair has a limit, an oscillating pair has none (theorem 32(4)).

approach	orders on $L \times A$	surviving sources	conditioned limit	checks
common order	several $m_l = m$	all of least order	weights $c_l / \sum c_j$	OR3
unequal order	some $m_l > m$	least order only	others vanish like $s^{m_l - m}$	OR3
one-sided mismatch	$(m^-, c^-) \neq (m^+, c^+)$	may differ by side	jump iff the leading conditionals differ	OR4
analytic passage	even, equal on both sides	same on both sides	removable	SJ5, OR4
flat, Hardy	beyond all powers	least order	exists	OR5
flat, not Hardy	not comparable	undefined	may not exist	OR5

Limits of Step 3. Orders are taken along one-parameter paths; several parameters are Step 4a–4b (section 17). Decisions taken on conditioned laws at the wall, whose acts depend on the leading coefficients and, on ties, on the next order, are Step 4c (theorem 46). The graded classes of maps planned for Step 3 in the earlier draft move to Step 4, with the contextual second-order theorem.

17 Step 4a: several parameters at once

In the lab, several things move together: the share of a source, the rate at which it produces an event, the size of a batch. A one-parameter path is a choice of how they move together, and the data do not make that choice. This section lets d parameters $\varepsilon = (\varepsilon_1, \dots, \varepsilon_d)$ tend to 0 independently, in the open orthant. Its message, shown by five small examples, is that the order of section 16 becomes a *fan*: a finite division of the possible rates into cones, each with its own winning sources.

Setting. Each cell of Λ_ε on $L \times A$ is a polynomial $f_c(\varepsilon) = \sum_\alpha a_{c,\alpha} \varepsilon^\alpha$, non-negative on the orthant near 0, and $\Phi(\varepsilon) = \mathfrak{C}_A(\Lambda_\varepsilon)$ is the conditioned state. A *monomial arc* is $\varepsilon_i = c_i s^{w_i}$ with weights $w \in \mathbb{Q}^d_{>0}$ (how fast each parameter vanishes) and coefficients $c \in \mathbb{R}^d_{>0}$, and $s \downarrow 0$. A *Puiseux arc* has $\varepsilon_i = c_i s^{w_i}(1 + o(1))$. The *tropical order* of a cell along w is $\operatorname{trop}_w(f) = \min\{\langle w, \alpha \rangle : a_\alpha \neq 0\}$, and its *initial form* $\operatorname{in}_w f$ collects the monomials that attain it. The *Newton polyhedron* is $N = \operatorname{conv}\big(\bigcup_c \operatorname{supp} f_c\big) + \mathbb{R}^d_{\geq 0}$.

Proposition 35 (limits along arcs)**.** *Along a Puiseux arc with data (w, c), $f_c(\varepsilon(s)) = \operatorname{in}_w f_c(c)\, s^{\operatorname{trop}_w f_c} + o(s^{\operatorname{trop}_w f_c})$, and $\operatorname{in}_w f_c(c) \geq 0$. If the initial forms of the cells of least tropical order do not all vanish at c (the arc is* non-degenerate*), the conditioned state tends to the normalised vector of those initial forms evaluated at c. Cells whose initial form vanishes at c simply drop out of the limit.*

Proof. Each monomial contributes $a_\alpha c^\alpha s^{\langle w, \alpha \rangle}(1 + o(1))$. The leading part is non-negative because $f_c \geq 0$. Then apply theorem 32 to the one-parameter germs. MP1 checks this exactly on 300 random families with $d = 2, 3$, and against evaluation at $s = 10^{-6}$. □

Theorem 36 (the order fan and the limit set)**.** *(1)* The fan. *For $w > 0$, the cells of least order are those with a monomial on the face F_w of N that minimises $\langle w, \cdot \rangle$, and $\operatorname{in}_w f_c$ is the part of f_c on that face. The face is constant on the relatively open cones of the normal fan of N, and there is one cone for each bounded face. This finite fan is the* order fan*.*

(2) The limit set. *Over non-degenerate arcs, the set of limits is the union over the bounded faces F of $\{\text{normalise}(f_c^F(c))_c : c > 0\}$, where f_c^F is the part of f_c on F. A vertex contributes one point. An edge or a higher face contributes a connected set swept out as c varies.*

(3) A sufficient condition. *If N has a single vertex α^*, that is, if the exponents have a componentwise minimum, then $\Phi(\varepsilon)$ converges as $\varepsilon \to 0$ in the orthant, along any route, to the normalised vector $(a_{c,\alpha^*})_c$.*

(4) Monomial cells. *If every cell is a single monomial, the limit exists iff the exponents have a componentwise minimum.*

(5) Degenerate directions. *If the initial forms of* all *least-order cells vanish at some $c > 0$ (equivalently, the initial form of $P(A) = \sum_c f_c$ does), the order along those arcs is higher than* trop_w *and depends on the next terms of the arc. There the fan must be refined, recursively (theorem 59).*

For semialgebraic families, the curve selection lemma [23] says that the limit exists iff it exists, and is the same, along every semialgebraic arc. Such arcs have Puiseux expansions, so (1)–(5) cover every route once degenerate directions are refined. Lines through 0 are not enough (example 39).

Proof. (1) $\langle w, \cdot\rangle$ attains its minimum on N on a face, which is bounded because $w > 0$. Faces of a polyhedron and their normal cones are finitely many. (2) This is proposition 35 cone by cone. (3) For $\alpha \in \mathrm{supp}$, $\alpha \neq \alpha^*$, the difference $\alpha - \alpha^*$ is ≥ 0 and non-zero, so $\varepsilon^\alpha/\varepsilon^{\alpha^*} \leq \max_i \varepsilon_i^{\min(\alpha-\alpha^*)>0} \to 0$ uniformly. So every cell is $a_{c,\alpha^*}\varepsilon^{\alpha^*}(1+o(1))$. The coefficients are ≥ 0 because the cells are, and not all zero. (4) If there is no componentwise minimum, N has two vertices $\alpha \neq \beta$. Their cones give limits supported on the cells with exponent α and on those with exponent β, which are disjoint sets. (5) is example 40. MP2 checks (1) by scanning weights (V vertices and $V-1$ bounded edges in the plane), and MP3 checks (3)–(4) on 300 monomial families. Step 4f proves (5) as a recursive theorem. □

Example 37 (the race)**.** Source 1 has a small share t and produces the event at rate s; source 2 produces it at rate s^2. So $\Lambda(1, A) = ts$ and $\Lambda(2, A) = (1-t)s^2$. The exponents $(1, 1)$ and $(2, 0)$ have no componentwise minimum. The fan has two cones and a ray (fig. 9(a)–(b)). Along $t = cs^w$:

$$\pi_1^A \to \begin{cases} 1 & w < 1 \text{ (the share vanishes more slowly than the event rate)}, \\ \frac{c}{1+c} & w = 1, \\ 0 & w > 1. \end{cases}$$

The limit set is all of $[0, 1]$ (fig. 9(c)). Which source explains a rare event depends on a race between how rare the source is and how rare the event is within it. A one-parameter analysis hides this, because choosing the path decides the race.

Example 38 (dominance)**.** $\Lambda(1, A) = st$ and $\Lambda(2, A) = st + s^2t$. The exponent $(1, 1)$ is a componentwise minimum, shared by both sources, so $\pi^A \to (\frac{1}{2}, \frac{1}{2})$ by every route (theorem 36(3)). Both sources reach the event at the same order in every parameter, so no race takes place.

Example 39 (lines are not enough)**.** Cells x^2 (source 1) and y (source 2). Along every line $y = vx$ the cell y wins, and the share of source 1 tends to 0. Along the parabola $y = cx^2$ it tends to $1/(1+c)$, so there is no limit (fig. 9(d)). The weight $w = (1, 2)$ is the edge normal of N, and no line has it.

Example 40 (a degenerate direction)**.** Source 1 sees the event only through the gap between two rates, $\Lambda(1, A) = (x-y)^2$; source 2 through $\Lambda(2, A) = x^3$. Along $y = cx$ with $c \neq 1$, source 1 wins (order 2 against 3). On the line $c = 1$ its initial form vanishes. Along the refined arc $y = x + bx^{3/2}$, the share of source 1 tends to $b^2/(1+b^2)$. A whole interval of limits is hidden inside one direction of the fan.

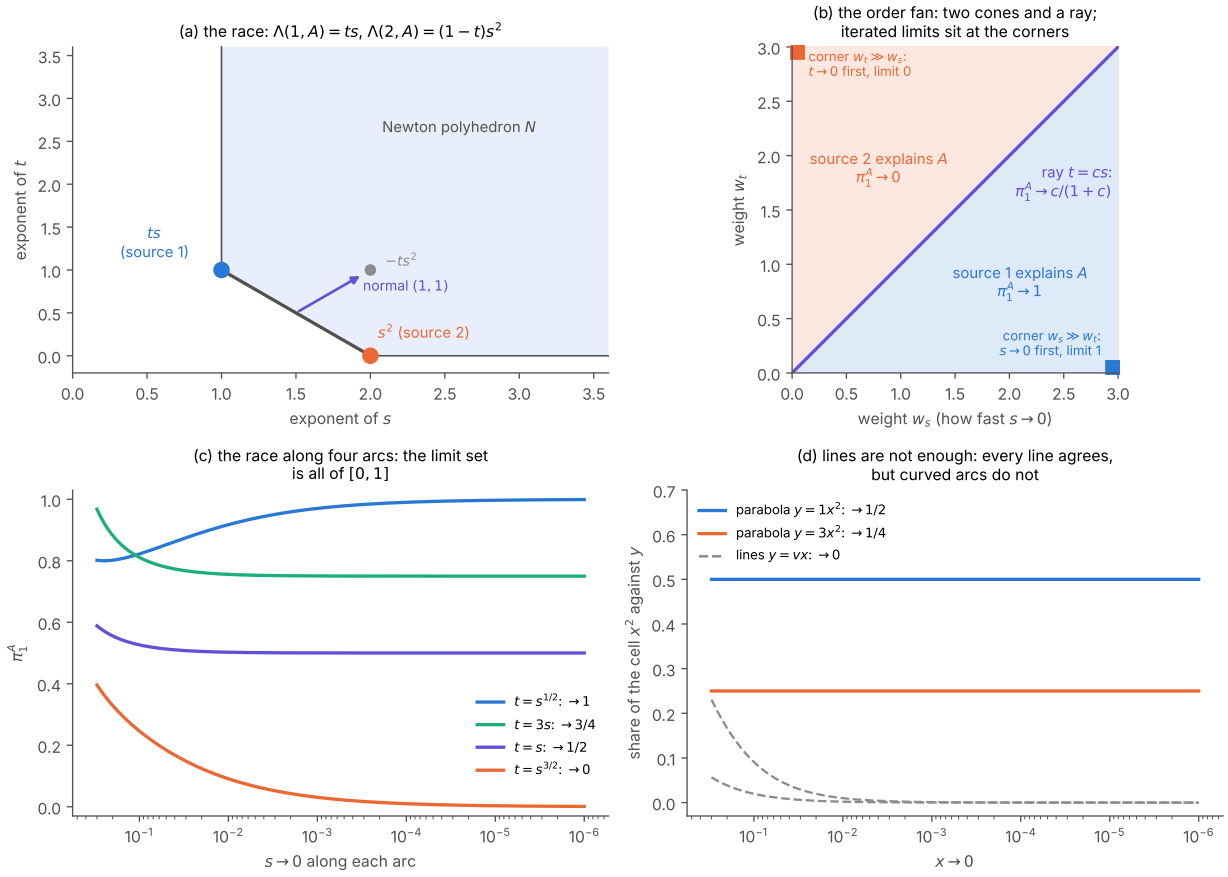


Figure 9: Several parameters. (a) The Newton polyhedron of the race (example 37). (b) Its order fan: in each cone one source explains A; on the ray both do, in proportion $c : 1$; the iterated limits sit at the two corners (theorem 41). (c) The race along four arcs, with four different limits. (d) Example 39: every line gives 0, every parabola something else.

18 Step 4b: iterated limits and the tangent tower

Sending the parameters to 0 one after another is an extreme way of moving through the fan: each parameter is infinitely smaller than the next. This is exactly how the Weil algebra of Step 2 orders its generators, $e_1 \ll e_2 \ll \dots$. So the tangent tower computes iterated limits, the corners of the fan, and the fan fills in everything between them.

Theorem 41 (iterated limits are corners of the fan)**.** *Fix an order $(i_1, \dots, i_d)$ and send $\varepsilon_{i_1} \to 0$ with the others fixed, then ε_{i_2}, and so on. Each stage has a limit (for the parameters still held fixed, away from the zero set of a non-zero polynomial), and the result is the normalised vector of coefficients at the lexicographic minimum α^σ of the exponents: minimise α_{i_1}, then α_{i_2}, and so on. This α^σ is a vertex of N. The result equals the limit along any weight in the corner cone $w_{i_1} \gg w_{i_2} \gg \cdots \gg w_{i_d}$. Different orders can give different limits. If they all agree, the full limit can still fail to exist.*

Proof. At the first stage every cell is a polynomial in ε_{i_1} with coefficients in the other parameters. Its lowest power wins, with a non-negative coefficient polynomial. Repeat. A lexicographic minimum minimises $\langle w, \cdot \rangle$ for sufficiently separated weights, so it is a vertex. For the last claim, take $x^3 + y^3$ (source 1) against xy (source 2). Both iterated limits give source 1, yet along $x = y$ source 2 wins: the middle vertex $(1, 1)$ of N is seen by no corner. MP5 checks this exactly on 300 families, all orders, against the arc limit for super-increasing weights and against nested numerical evaluation. □

Proposition 42 (the tower computes the corners)**.** *Let every cell be multilinear: exponents in $\{0, 1\}^d$, as for a product $\pi_l(t)\, p_{l,s}(x)$ of a share and a likelihood, each moving in its own parameter.*

(1) The Weil point $\Lambda(e_1, \dots, e_d)$ lies in $T^d \Delta_{L \times X}$.

(2) The first monomial e^S, in the order of the lexicographic lemma, that carries mass on $L \times A$ gives, normalised, the iterated limit that sends ε_1 to 0 first, then ε_2, and so on.

(3) The canonical flip c of the tangent structure, which exchanges e_1 and e_2, gives the iterated limit in the other order.

So the tangent tower sees exactly the corners of the order fan, and its flip moves between them. The cones in between, such as the ray $t = cs$ of the race, are invisible to it. They need the fan.

Proof. (1) For each set U of generators, setting the other parameters to 0 keeps a cell non-negative. Along a super-increasing arc, its leading coefficient is the first non-zero one in the lemma's order, so that coefficient is positive. (2) In the lemma's order e^S precedes $e^{S'}$ iff the smallest generator of $S \triangle S'$ lies in S'. So monomials without e_1 come first, then, among those, monomials without e_2, and so on: the lexicographic minimum of theorem 41 with ε_1 first. (3) The flip is the substitution exchanging e_1 and e_2, that is, exchanging the roles of ε_1 and ε_2. MP6 checks (1)–(3) on 200 random multilinear families, 95 of them with the flip. □

Example 43 (two sources in the tower)**.** Source 1's share is ε_1 and source 2's likelihood of A is ε_2. On $L \times A$ the Weil point has source 1 on e_1 and source 2 on e_2. The lemma's order is $(1, e_2, e_1, e_1 e_2)$. The first term with mass on A is e_2, so source 2 explains A: sending the share ε_1 to 0 first removes source 1. The flip puts e_1 first, and source 1 explains A. On the ray $\varepsilon_1 = c\,\varepsilon_2$, between the corners, the weights tend to $(\frac{c}{1+c}, \frac{1}{1+c})$, which no point of T^2 records.

example	cells on $L \times A$	componentwise min	corners (iterated limits)	full limit	what it shows
37 race	ts, $(1-t)s^2$	none	0 and 1	none; limit set $[0,1]$	a race between rarities
38 dominance	st, $st+s^2t$	$(1,1)$	$\frac{1}{2}$, $\frac{1}{2}$	$(\frac{1}{2},\frac{1}{2})$	no race
39 curved path	x^2, y	none	0 and 1	none, though every line gives 0	arcs, not lines
40 degenerate	$(x-y)^2$, x^3	none	1 and 0	none; an interval inside one direction	refine the fan
theorem 41	x^3+y^3, xy	none	1 and 1	none	corners are not enough
43 tower	ε_1, ε_2	none	0 and 1 (the flip)	none	the tower sees corners only

What this sets up. The cross term $\sum_l d\pi_l\, dp_l$ of a pooled law, the target of Step 4d, is the coefficient of e_1e_2 when shares move along e_1 and sources along e_2. It is a mixed order, read at the last place of the lemma's order. Step 4c will attach decisions to the cones of the fan.

19 Step 4c: parameters as an object, and decisions on the fan

So far the parameters of Steps 3–4b were coordinates: a family was a map from a parameter space into the labelled simplex, and arcs were chosen by hand. This section gives parameters their own place in the categorical picture, separate from both kinds of context already in use. A *measurement context* is a window of jointly observed variables, and a *source* is the discrete label carried by W. A *parameter* is a quantity that moves and that drives a transition: a share, a rate, a design choice. It lives in a parameter space, and changing how the parameters are described is a 2-cell. Decisions are then attached to the order fan, and they transport along those 2-cells.

Definition 44 (parameterised source-indexed kernels)**.** The 2-category Para_L has the finite sets as objects.

- A morphism $X \to Y$ is a pair (P, f): a parameter space P (here a semialgebraic subset of some $\mathbb{R}^d$) and a map $f : P \times L \times X \to \mathcal{D}Y$. For each parameter value it is a source-indexed kernel (proposition 4).
- Composition is $(Q,g)\circ(P,f) = (P\times Q,\ (p,q,l,x)\mapsto \sum_y f(p,l,x)(y)\, g(q,l,y))$: parameters accumulate, and one source is held throughout.
- A 2-cell $(P',f') \Rightarrow (P,f)$ is a *reparametrisation*, a map $r : P' \to P$ with $f' = f \circ (r \times \mathrm{id})$.

This is the Para construction [15, 16] applied to the two-sided Kleisli category of λ. A parameterised labelled law is a morphism (P, Λ) from the one-point set into $L \times X$.

Proposition 45 (the layers act by post-composition)**.** *(1)* Para_L *is a 2-category, strict once* $(P \times Q) \times R$ *is identified with* $P \times (Q \times R)$*. Reparametrisations whisker:* $(g \circ f) \circ (r \times r') = (g \circ r') \circ (f \circ r)$*.*

(2) Labelling, pooling, marginalisation and forgetting provenance are parameter-free morphisms. Applying them after a reparametrisation equals reparametrising after applying them.

(3) Conditioning is partial. Its domain for (P, Λ) *is* $\{p : \Lambda_p(L \times A) > 0\}$*, and that domain pulls back along every reparametrisation.*

So reparametrisation commutes with every layer operation at every parameter value. What it does not preserve is behaviour at the boundary *of the parameter space. That behaviour is what the order fan describes, and reparametrisation can change which part of the fan is reachable (example 48).*

Proof. (1) Composition of kernels with a fixed source is associative (proposition 4), and the parameters ride along as a product. Whiskering is substitution. (2) Post-composition with a parameter-free morphism commutes with precomposition by r. (3) The domain condition is evaluated pointwise. PR1 and PR2 check (1)–(3) exactly on random parameterised kernels and laws. □

Arcs are 2-cells; the germ field. An arc into the corner of P is a reparametrisation from the one-parameter germ line, $s \mapsto \varepsilon(s)$. So Steps 3–4b classify such 2-cells by what happens at $s = 0$. The same information sits in one algebraic object. Let $\mathbb{K} = \mathbb{R}\{\{s\}\}$ be the field of real Puiseux series. It is a real closed ordered field, with valuation v (the least exponent) and a standard part st (the constant term of a bounded series). A Puiseux arc is a point $\varepsilon \in \mathbb{K}^d_{>0}$ with $v(\varepsilon_i) = w_i$ and leading coefficients c_i. Evaluating the cells there gives a labelled law over $\mathbb{K}$. Its valuations are the tropical orders $\mathrm{trop}_w f_c$ (off degenerate directions), and the standard part of its conditioned state is the arc limit of proposition 35 (PR3). Sending parameters to 0 one after another corresponds to nesting the fields, $\mathbb{R}\{\{s_1\}\}\{\{s_2\}\}\cdots$. These are the corners of the fan (theorem 41). For multilinear cells no squares occur, so the Weil algebra of Step 2 already suffices (proposition 42).

Decisions. A decision problem is a finite set of acts with utilities u_a on $L \times X$. The act chosen at a parameter value maximises $\langle u_a, \Phi(\varepsilon)\rangle$, the expected utility under the conditioned state. Along an arc, the comparison of a and b is the germ $D_{ab}(s) = \langle u_a - u_b, \Phi(\varepsilon(s))\rangle$, an element of $\mathbb{K}$.

Theorem 46 (the decision fan)**.** *(1) Along every Puiseux arc the optimal act is eventually constant. It is decided in the ordered field $\mathbb{K}$ by the sign of the leading coefficient of each D_{ab}. A comparison is undecided only if $D_{ab} \equiv 0$, a genuine tie along the whole arc.*

(2) If the limiting state $\mathrm{st}\,\Phi$ has a unique optimal act, the eventual act is that act. If the limiting state has a tie, the next order of D_{ab} breaks it.

(3) On a cone of the order fan that belongs to a vertex of N, the limiting state is fixed, so the limiting act is constant there. A cone that belongs to an edge or a higher face is cut further by the walls $\langle u_a - u_b, \mathrm{in}_F f(c)\rangle = 0$, which are polynomial in the arc coefficients c.

(4) Transport. *Let $r(\delta) = \delta^R$ be a monomial reparametrisation, $\varepsilon_i = \prod_j \delta_j^{R_{ij}}$. Then the arc (w', c') in the new parameters maps to the arc (Rw', c'^R) in the old ones. Arc limits and eventual acts are pulled back: the order fan and the decision fan of $\Lambda \circ r$ are the preimages of those of Λ under $w' \mapsto Rw'$.*

Proof. (1) Each D_{ab} is a ratio of Puiseux series in s with a positive denominator, so it is 0 or eventually of the sign of its leading coefficient. Finitely many comparisons decide the act. (2) $\mathrm{st}\, D_{ab}$ is the limiting comparison; if it is 0, the leading coefficient lies at a higher order. (3) This is theorem 36(2) composed with a linear functional. (4) $\delta_j = c'_j s^{w'_j}$ gives $\varepsilon_i = \prod_j c_j'^{R_{ij}} s^{\sum_j R_{ij} w'_j}$; then apply proposition 45(2). PR4 checks (1)–(2) on 300 random families and utilities, including 49 ties at the limit broken at higher order, against evaluation at $s = 10^{-8}$. PR6 checks (4) on 200 random reparametrisations. □

Example 47 (which source to audit)**.** In the race of example 37, a flagged record arrives and one source is to be audited. Auditing the source that produced the record pays 1; auditing the other pays 0. The optimal act audits the source with the larger posterior weight. The decision fan (fig. 10(a)) is:

- when the share vanishes more slowly than the event rate ($w_t < w_s$), audit the small trials (source 1);
- when it vanishes faster ($w_t > w_s$), audit the registry (source 2);
- on the ray $t = cs$, audit source 1 iff $c > 1$, and source 2 iff $c < 1$;

- at $c = 1$ the limiting state is $(\frac{1}{2}, \frac{1}{2})$, a tie. The comparison germ is $D_{12}(s) = s^3/(\text{positive})$, from the factor $1 - t$ in the registry's cell, so the next order decides: audit source 1 (PR5).

The tie at the limit is broken by a term that no first-order analysis sees. This is a third-order effect deciding a first-order question.

Example 48 (two study designs are two reparametrisations)**.** The same two sources can be studied under different designs. Each design is a reparametrisation of (s, t), a 2-cell into the race.

- *Design 1*, $t = s\,u$: the small-trial share is set in proportion to the event rate. The weights become $w = (w_s,\ w_s + w_u)$, which is exactly the cone $w_t > w_s$. The exponents of $\Lambda \circ r$ have the componentwise minimum $(2, 0)$, so the limit exists. By every route to the corner, the registry explains a flagged record: audit source 2.
- *Design 2*, $s = u\,v$ and $t = u$: the event rate is tied to the share. The weights become $w = (w_u + w_v,\ w_u)$, the cone $w_s > w_t$. The componentwise minimum is $(2, 1)$, the limit exists, and the small trials explain the record: audit source 1.

In the raw parameters the race has no limit at all (example 37). Each design fixes the race in advance by occupying one cone of the fan, and theorem 46(4) transports the decision along the 2-cell (fig. 10; PR6). A design that sits on the ray, such as $t = s\,(1 + u)$, ends in the tie $(\frac{1}{2}, \frac{1}{2})$. There the comparison $ts - (1 - t)s^2 = s^2u + s^3(1 + u)$ is positive, so it audits source 1 by every route.

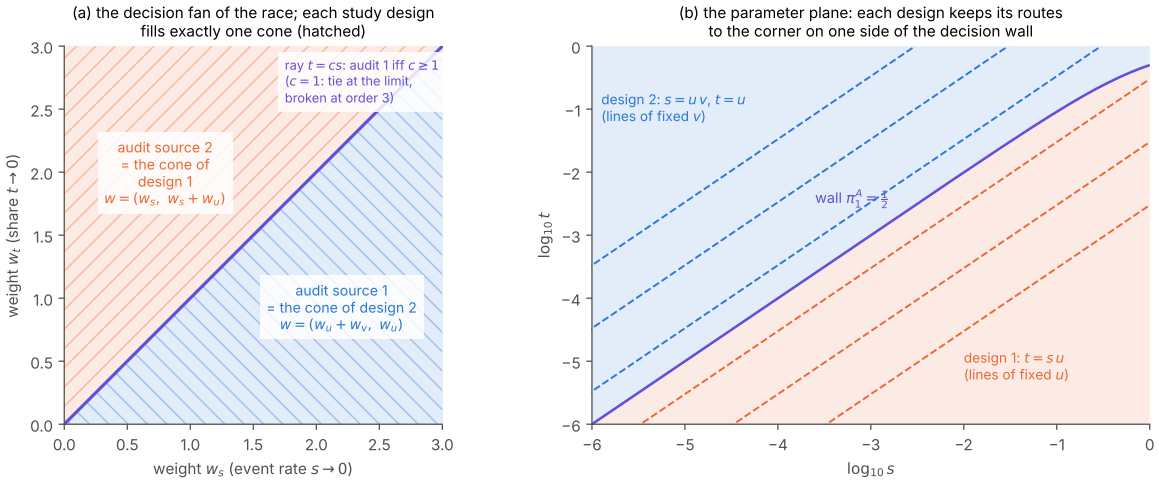


Figure 10: Step 4c on the race. (a) The decision fan of example 47. Each design of example 48 fills one cone exactly. (b) The parameter plane on log scales, with the decision wall $\pi_1^A = \frac{1}{2}$. The routes of design 1 (lines of fixed u) and design 2 (lines of fixed v) stay on opposite sides.

What this sets up for Step 4d. The cross term $\sum_l d\pi_l\, dp_l$ is the coefficient of $e_1 e_2$ when the shares move along one parameter and the sources along another. Here that means the mixed order of the comparison germ on the diagonal cone of the share–likelihood fan. The graded classes of maps and the contextual second-order theorem will be stated in this setting.

20 Step 4d: the decorated fan, the cross term, and Theorem J

This step does three things.

- It separates the two levels of information in the order fan, in the probability-decorated order fan: the geometry, which says which exponents can dominate, and the probability carried on each cone, which says which sources survive and with what weights.
- It isolates the cross term $\sum_l a_l\, u_l$ of pooling exactly.
- It derives the band of the lab's Theorem J [1, Theorem J] as an inequality between orders on a *statistical* fan, in which the sample size is one more parameter, and it marks the cones where J's Gaussian bands are valid and where they are not.

20.1 The probability-decorated order fan

Definition 49 (probability-decorated order fan)**.** A *probability-decorated order fan* consists of a geometric Newton fan together with compatible facewise initial-form data, which determine the limiting conditioned laws. For cells (f_c) on $L \times A$ it is the pair $(\Sigma_N, \mathcal{O})$:

- Σ_N is the geometric fan of theorem 36, with one cone σ_F for each bounded face F of N;
- $\mathcal{O}$ is the *facewise presheaf of initial forms*. It is the functor on the poset of bounded faces of N, ordered by inclusion, that sends F to the tuple of face parts $(f_c^F)_c$, the monomials of each cell with exponents in F. A face $G \subseteq F$ gets the restriction $\rho_{F,G} : f^F \mapsto (f^F)^G$.

The decoration on F determines the *limit map* $\ell_F(c) = \text{normalise}\big(f_c^F(c)\big)_c$ on the cone of F. A *decision decoration* adds the comparison functionals $\langle u_a - u_b, \ell_F(c)\rangle$ of theorem 46.

Proposition 50 (the facewise presheaf of initial forms)**.** *(1) The restrictions compose,* $(f^F)^G = f^G$ *for* $G \subseteq F$*, so* $\mathcal{O}$ *is a presheaf on the face poset. In terms of weights,* $\text{in}_{w'}(\text{in}_w f) = \text{in}_{w+\epsilon w'} f$ *for all small* $\epsilon > 0$*.*

(2) Along an edge $F = [\alpha, \beta]$*, if no initial form vanishes, the limit map tends to the vertex limit* ℓ_α *as* $c^{\beta-\alpha} \to 0$ *and to* ℓ_β *as* $c^{\beta-\alpha} \to \infty$*. So the limit set over an edge is a path joining the limits of its two vertices.*

(3) The geometric fan does not determine the limits. Two decorations of the same Newton polyhedron can have different limit sets.

Proof. (1) The monomials of f that minimise $\langle w, \cdot\rangle$ and then, among those, $\langle w', \cdot\rangle$ are those that minimise $\langle w + \epsilon w', \cdot\rangle$ for small ϵ. (2) On F every exponent is $\alpha + t(\beta - \alpha)$ with $t \in [0, 1]$, so $f^F(c) = c^\alpha \sum_t a_t (c^{\beta-\alpha})^t$, and the lowest or highest t dominates. (3) Take cells x (source 1) and y (source 2), against $x + y$ (source 1) and x (source 2). Both have the Newton polyhedron with vertices $(1, 0)$ and $(0, 1)$. Along $y = cx$, the first gives $\pi_1^A = \frac{1}{1+c}$, filling $[0, 1]$. The second gives $\pi_1^A = \frac{1+c}{2+c}$, filling only $[\frac{1}{2}, 1]$ (fig. 11(d)). ST1 checks (1)–(2) exactly on random families and 53 edges; ST2 checks (3). □

So the fan says which exponents can dominate, and the decoration says which sources survive and with what weights. Only the pair determines the limiting conditioned law.

Remark 51 (why "presheaf" and not "sheaf")**.** Proposition 50(1) gives a coherent restriction system, a presheaf, and nothing in this paper uses a gluing property. A sheaf needs a topology and gluing. One can supply both: give the face poset its Alexandrov topology, whose open sets are the sets of faces closed under passing to subfaces. On a finite poset, presheaves for this topology extend uniquely to sheaves, the *cellular sheaves* of [17], and gluing and uniqueness then hold automatically. Moreover $\mathcal{O}$ has a canonical global section, the cells themselves, from which every local section is obtained by restriction. We keep the safer name, *facewise presheaf*, because the content of the proposition is (3), not gluing.

20.2 The cross term of pooling

Proposition 52 (the cross term)**.** *(1) Pooling is exactly bilinear:* $\mu_n(\pi + a, p + u) - \mu_n(\pi, p) = \sum_l a_l p_l + \sum_l \pi_l u_l + \sum_l a_l u_l$.

(2) With the shares moving along e_1 *and the sources along* e_2*, the coefficient of* $e_1 e_2$ *is the cross term* $\sum_l a_l u_l$*. The second-order coefficients of a move along shares alone, or along sources alone, vanish. The cross term is the whole curvature of pooling.*

(3) After conditioning, the mixed second derivative of $C_A \circ \mu_n$ *goes through the Bayes weights:*

$$\partial_a \partial_u (C_A \circ \mu_n) = \sum_l \partial_a \partial_u \pi_l^A \, C_A(p_l) + \sum_l \partial_a \pi_l^A \, DC_A(p_l)[u_l].$$

(4) On a fan where the shares move at order w_a *and the sources at order* w_u*, the cross term has order* $w_a + w_u$ *(theorem 30).*

Proof. (1) Expand. (2) is (1) in W_2. (3) Differentiate theorem 11; $C_A(p_l)$ does not depend on a. (4) A product of germs. ST3 checks (1)–(3), (1) exactly and (2)–(3) in the Weil algebra against the explicit formula. □

20.3 Theorem J on a statistical fan

Statistical assumptions. The statements of this subsection and the next hold under the following assumptions, which are those of J [1, Theorem J] made explicit.

(S1) Records are independent. The share π of source C and the genuine law are fixed by design, not estimated.

(S2) *Gaussian model* (theorem 53). Each source statistic is normal with known variance σ^2, exactly or as the large-count limit. This requires the relevant counts to grow, that is, to lie in the Gaussian cone of proposition 54.

(S3) *Poisson model* (theorem 55). Flags, the records falling in a rare event A, arrive as independent Poisson counts. This is exact for a Poisson process of records, and it is the rare-event limit of binomial counts.

(S4) The adversary is deterministic: it chooses its share and its shift (or its flag rate) in advance, and only raises them.

(S5) The exposure test and its level are fixed before the data. Arcs are taken along $N \to \infty$, with every parameter a germ in a Hardy field (in practice, a power of $s = N^{-1/2}$).

Recall the setting of J [1, Theorem J], which uses a Gaussian model:

- source C holds a share π of N records, and its records' statistic is shifted by D;
- its source statistic has mean $a = D\sqrt{\pi(1-\pi)N}/\sigma$;
- the pooled estimate moves by πD, which is $I = \pi D\sqrt{N}/\sigma$ pooled standard errors.

Pooling with a new source is a move from the corner $\pi = 0$ of the share simplex, so the impact πD is precisely the cross term of proposition 52: share times shift. Now treat the sample size as a parameter of Para_L: $s = N^{-1/2}$, the share $\pi \sim p\, s^{w_\pi}$, and the shift $D \sim d\, s^{w_D}$.

Theorem 53 (Theorem J as an inequality of orders)**.** *Along any such arc:*

(1) $I = a\sqrt{\pi/(1-\pi)}$ *exactly. The orders are* $v(a) = w_D + w_\pi/2 - 1$ *and* $v(I) = w_\pi + w_D - 1$. *The impact counts the share at full weight, and the exposure only at half weight.*

(2) The regimes are cones. *Below the exposure line* $w_D + w_\pi/2 = 1$, $a \to \infty$ *and* C *is exposed with probability tending to* 1. *Above it,* $a \to 0$ *and the exposure rate falls to the false-alarm rate. On the line, the exposure probability tends to* $1 - \Phi(c_s - a_0)$, *with* a_0 *the leading coefficient of* a.

(3) The band. *Bounded exposure,* $P(\text{exposed}) \le \beta$, *means* $a \le a_{\max} = c_s - \Phi^{-1}(1-\beta)$. *That forces* $w_D + w_\pi/2 \ge 1$, *hence* $v(I) \ge w_\pi/2 \ge 0$, *and*

$$I \le a_{\max}\sqrt{\pi/(1-\pi)} = B.$$

The impact stays of order one only at the single point $(w_\pi, w_D) = (0, 1)$*: a fixed share, with a shift shrinking like* $N^{-1/2}$. *That is J(2). If the share also vanishes* ($w_\pi > 0$), *the impact vanishes like* $s^{w_\pi/2}$.

(4) No tilt. *Under bounded exposure the raw pooled shift* πD *has order* $w_\pi + w_D \ge 1 + w_\pi/2 > 0$, *so it tends to* 0 *and cannot cross a fixed settling radius. The adversary that does tilt the pooled decision, with fixed share and fixed shift* ($w_\pi = w_D = 0$), *lies in the exposed cone. This is the limiting content of J(3).*

Proof. (1) J's identity [1, proof of Theorem J(2)], with v applied through theorem 30. (2) The source statistic is normal with mean a and variance 1. (3) Monotonicity of Φ, then (1). (4) A product of germs. ST4 checks the orders as slopes on log scales and the bound at three sample sizes and three shares. ST5 checks (2) by Monte Carlo along three arcs at $N = 10^3, 10^4, 10^5$: the exposure rate rises to 1, stays at $1 - \Phi(c_s - a_0) \approx 0.095$ on the line, and falls to the false-alarm rate 0.023. □

So J's band is the leading coefficient of the cross term on the one ray of the statistical fan where exposure is of order one. The band exists because exposure sees the share at half weight ($\sqrt{\pi}$), while the impact sees it at full weight (fig. 11(a)). This answers the question left open in the scope of Steps 1–3, for J's Gaussian model.

20.4 Where Gaussian bands are valid

J's bands are Gaussian, and Gaussian approximations need counts that grow. The order fan says when they do. Let an event have probability $P(A) \sim c\, s^{w_A}$, again with $s = N^{-1/2}$. Its expected count is $c\, s^{w_A - 2}$.

Proposition 54 (the Gaussian cone, the Poisson ray, and the empirical wall)**.** *(1) If $w_A < 2$, the count grows, conditional estimates such as $\hat{\pi}_l^A$ are asymptotically normal, and Wald bands are asymptotically valid.*

(2) If $w_A = 2$, the count tends to Poisson(c)*. There is no event with probability e^{-c}, and conditional estimates have a discrete limiting law, so Gaussian bands are not valid. In ST6, with $c = 3$ and $\pi_1^A = 0.3$, the nominal 95% interval covers about 50% at every N.*

(3) If $w_A > 2$, there is no event with probability tending to 1. The empirical conditional is undefined: an empirical wall*, the sample-level analogue of proposition 10.*

The same split applies to source C's own record count πN. J's Gaussian derivation needs $w_\pi < 2$. On the ray $w_\pi = 2$, C contributes a bounded number of records, and its source statistic is no longer normal.

Proof. The law of rare events for $w_A = 2$, the central limit theorem when the count grows, and $P(\text{no event}) = (1 - P(A))^N$. ST6 simulates all three cases at $N = 10^2, 10^4, 10^6$. ST7 is the negative control: the same procedure at a fixed $P(A) = 0.2$ covers within Monte Carlo error once $N \geq 10^4$. At $N = 100$ the Wald interval already under-covers (about 0.92), a known small-sample effect. □

20.5 A band on the Poisson ray

On the Poisson ray the source statistic is a count, and J's normal approximation fails. An exact discrete band replaces it. Under (S3), let C produce $X_C \sim \text{Poisson}(\mu_C)$ flags and the genuine sources $X_O \sim \text{Poisson}(\mu_O)$. Under the null, C flags at the genuine rate, so $\mu_C = \mu_0 = \pi\mu_O/(1-\pi)$. Use the *exact conditional test*: given $T = X_C + X_O$, under the null $X_C \sim \text{Binomial}(T, \pi)$, and the test rejects when $X_C \geq k_\alpha(T)$. Here $k_\alpha(T)$ is the least k with $P(\text{Bin}(T,\pi) \geq k) \leq \alpha$, or $T + 1$ if there is none. The adversary's excess is $\Delta = \mu_C - \mu_0$ expected flags. In pooled standard errors this is $\Delta/\sqrt{Np}$, where $Np = \mu_O/(1-\pi)$ is the genuine pooled flag mean.

Theorem 55 (the band on the Poisson ray)**.** *(1)* Exact band. *$k_\alpha(T+1) \leq k_\alpha(T) + 1$, so the rejection region grows with X_C. The exposure probability is therefore increasing in μ_C, and it is at most α under the null. An adversary with exposure at most β adds at most*

$$\Delta_{\max}(\mu_O) = \mu_C^{\max} - \mu_0 \text{ expected flags}, \qquad B_{\text{pois}} = \Delta_{\max}/\sqrt{Np} \text{ pooled standard errors},$$

where $\mu_C^{\max}$ is the exposure-β point. This holds at every count, with no normal approximation.

(2) Gaussian limit. *As $\mu_O \to \infty$ (the Gaussian cone), $B_{\text{pois}} \to B = \big(\Phi^{-1}(1-\alpha) - \Phi^{-1}(1-\beta)\big)\sqrt{\pi/(1-\pi)}$. This is J's band.*

(3) The Poisson ray. *At bounded μ_O, B_{pois} is a fixed number above B. The test also has a* blind spot*: with $t_b = \lceil \log\alpha / \log\pi \rceil - 1$, it cannot reject at all when $T \leq t_b$, whatever the adversary does.*

(4) The wall side. *As* $\mu_O \to 0$, $\Delta_{\max} \to \lambda^*$, *where* $P(\text{Poisson}(\lambda^*) \geq t_b + 1) = \beta$. *The band in standard errors diverges, so standard errors are the wrong unit there, and counts are the right one. The adversary's expected share of the flags tends to* 1*: an exposure-bounded adversary* owns *the rare event, and by theorem 32 it explains it in the conditioned law.*

Proof. (1) $\text{Bin}(T, \pi) + \text{Bern}(\pi) \geq k + 1$ implies $\text{Bin}(T, \pi) \geq k$, so $k_\alpha(T+1) \leq k_\alpha(T) + 1$. Hence, for fixed X_O, the rejection set is upward closed in X_C, and X_C is stochastically increasing in μ_C. Under the null the conditional test has level α given every T. Exposure is computed exactly by Poisson splitting: $T \sim \text{Poisson}(\mu_C + \mu_O)$, and given T, $X_C \sim \text{Bin}(T, \mu_C/(\mu_C + \mu_O))$. (2) The conditional statistic $(X_C - T\pi)/\sqrt{T\pi(1-\pi)}$ has mean $\approx \Delta\sqrt{1-\pi}/\sqrt{Np\,\pi}$, so the impact is that mean times $\sqrt{\pi/(1-\pi)}$, as in theorem 53(1). Then use the central limit theorem. (3) $k_\alpha(T) = T + 1$ iff even $X_C = T$ is not rare enough, that is, iff $\pi^T > \alpha$, that is, iff $T \leq t_b$. (4) $X_O = 0$ with probability tending to 1. Then $T = X_C$, and the test rejects iff $X_C \geq t_b + 1$. Checks: PB1 checks (1) for $T < 3000$ and three values of μ_O. PB2 checks exposure $= \beta$ at the band, against 200,000 Monte Carlo draws. PB3 checks (2): $B_{\text{pois}} = 0.389, 0.283, 0.250, 0.238$ at $\mu_O = 10, 10^2, 10^3, 10^4$, against $B = 0.2325$. PB4 checks (3): $t_b = 3$, and at $\mu_O = 3$ the band is 0.636 standard errors, $2.74\,B$. PB5 checks (4): $\lambda^* = 1.366$ flags, with share 0.9993 at $\mu_O = 10^{-3}$. □

So on the Poisson ray an adversary can move the pooled estimate almost three times further, in standard errors, than J's Gaussian band allows. Beyond the ray, a bounded number of injected flags becomes the whole explanation of the event (fig. 12(a)). This closes the Poisson case under (S1), (S3)–(S5).

What is proved, and what is not. Under (S1), (S2), (S4) and (S5), the band, the vanishing-share extension and the no-tilt limit follow from the order calculus and the cross term. Under (S1), (S3)–(S5), the exact discrete band of theorem 55 covers the Poisson ray and the wall side. A band for decisions that depend on the decoration near the walls is open.

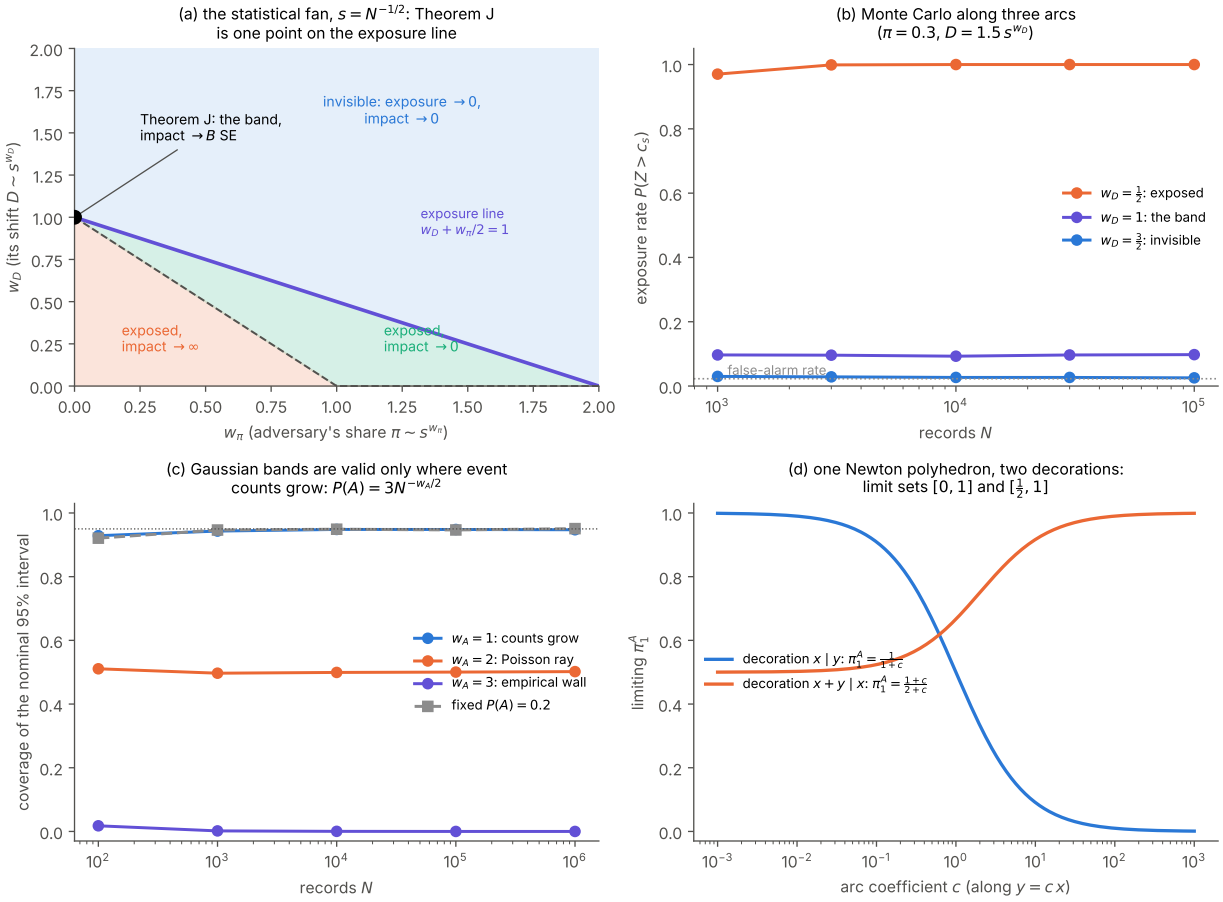


Figure 11: Step 4d. (a) The statistical fan in the share and shift orders, with $s = N^{-1/2}$. Theorem J is the point $(0, 1)$ on the exposure line (theorem 53). (b) Monte Carlo exposure rates along three arcs. (c) Coverage of nominal 95% Wald intervals: valid where counts grow, failing on the Poisson ray, and undefined beyond it (proposition 54). (d) One Newton polyhedron with two decorations (proposition 50(3)).

21 Step 4e: graded classes of maps

Steps 2–4 kept asking one question of each map: does it keep feasibility at order k? This section names the answer and shows that every layer operation passes at every order. It also locates exactly what fails: subtraction.

Definition 56 (graded classes)**.** For $1 \le k \le \infty$, $\mathcal{M}_k$ is the class of smooth maps f between products of simplices, partial maps on relatively open domains included, such that $T^k f$ maps T^k of the domain into T^k of the codomain at every point of the domain. ($\mathcal{M}_\infty = \bigcap_k \mathcal{M}_k$.) A map is *subtraction-free* if each output cell is built from the input coordinates and positive constants by sums, products, and division by expressions that are positive on the domain.

Theorem 57 (the graded classes)**.** *(1) Each $\mathcal{M}_k$ is a subcategory, and $\mathcal{M}_1 \supseteq \mathcal{M}_2 \supseteq \mathcal{M}_3 \supseteq \cdots \supseteq \mathcal{M}_\infty$.*

(2) The first two inclusions are strict:

- *$q \mapsto (1-(q_1-q_2)^2, (q_1-q_2)^2, 0)$ lies in $\mathcal{M}_1$ but not in $\mathcal{M}_2$;*
- *$q \mapsto (1-g, g, 0)$ with $g = x_1(x_1-x_2)^2$ lies in $\mathcal{M}_2$ but not in $\mathcal{M}_3$.*

(3) Every subtraction-free map lies in $\mathcal{M}_\infty$.

(4) Consequently every layer operation lies in $\mathcal{M}_\infty$ on its domain:

- *stochastic matrices, λ, $\mathcal{D}\epsilon$ and the marginalisations M_S;*
- *pooling μ_n, the labelled joint J and products of laws;*
- *conditioning C_A and the labelled conditioning $\mathfrak{C}_A$;*
- *the Bayes-weight map $(\pi, p) \mapsto \pi^A$ and the Bayesian inverse $K^\dagger$.*

In particular, conditioning keeps feasibility at every order. This settles the question left open in Step 2, which went only as far as order two.

Proof. (1) T^k is a functor. The zero section $T^k \to T^{k+1}$ (zero coefficients on monomials containing e_{k+1}) preserves and reflects membership, because in the lexicographic lemma it interleaves each sequence with zeros. So if $T^{k+1}f$ preserves T^{k+1}, then $T^k f$ preserves T^k. (2) The first map is the counterexample of [3, Proposition "morphisms"]. The second is the order-three example of [3, Proposition "order k: necessity, and order three"], which passes the second-order criterion. (3) $T^k f$ evaluates the same expression in the Weil algebra W_k. The input cells lie in $T^k H$. $T^k H$ is a convex cone, so it is closed under sums and positive multiples, and it is closed under products by theorem 20. A Weil number with positive constant term lies in $T^k H$ (every lexicographic sequence starts with a positive entry), and so does its inverse, whose constant term is the inverse of a positive number. So every output cell lies in $T^k H$. The output equations ($\sum_y f_y = 1$ and zero sums for the higher coefficients) hold because a rational identity on the domain holds in W_k wherever its denominators are invertible. (4) Each listed map is subtraction-free on its domain. For example, $C_A(p)(x) = p(x) \cdot \big(\sum_{x' \in A} p(x')\big)^{-1}$ with $p(A) > 0$, and $\pi_l^A = \pi_l p_l(A) / \sum_j \pi_j p_j(A)$. Checks: GC1 checks the zero section; GC2 and GC3 check the separations by direct T^2 and T^3 sampling. GC4 checks units and inverses for $k \le 4$. GC5 checks (3)–(4) on 2300 sampled points of T^3 and T^4 at corners, for conditioning, the labelled conditioning, the Bayes weights, the Bayesian inverse, products, and random normalised positive polynomial maps. □

Remark 58 (where feasibility breaks)**.** Theorem 57(3) says that, within the layers, only subtraction can take a map out of $\mathcal{M}_k$. Both separating examples square a contrast $q_1 - q_2$. GC6 finds that every one of 40 random maps carrying the square of a contrast leaves $\mathcal{M}_2$ at the vertex. In the lab, contrasts are exactly the estimators: covariances, tetrads, the decision gap $U_T - U_P$. This is why the curvature that produces the pooling paradox [3, Proposition "pooling paradox = curvature"] lives in the decision functional and never in a layer operation (fig. 12(b)). The converse fails: at relative-interior points every smooth map lies in every $\mathcal{M}_k$, so $\mathcal{M}_\infty$ is much larger than the subtraction-free maps. The graded classes are about boundaries.

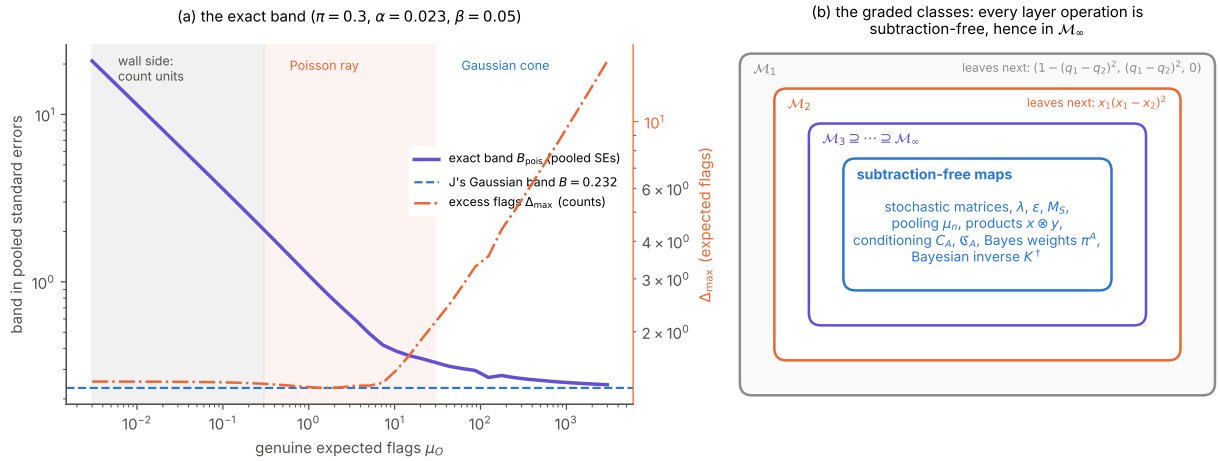


Figure 12: (a) The exact band of theorem 55 in pooled standard errors (left axis) and in expected flags (right axis). It falls to J's Gaussian band B in the Gaussian cone, stays well above it on the Poisson ray, and diverges in standard-error units on the wall side, where the count $\Delta_{\max} \to \lambda^* \approx 1.37$ is the right measure. (b) The graded classes (theorem 57).

Next. The degenerate directions of theorem 36(5) are resolved in Step 4f (theorem 59).

22 Step 4f: degenerate directions, recursively

Theorem 36(5) left one gap. Along a direction where the initial forms vanish, the leading terms say nothing, and the next terms of the arc decide. Example 40 showed a whole interval of limits hidden inside a single direction. This section shows how to resolve such directions, recursively and in finitely many steps, for two parameters.

The key simplification is this. A direction needs refining only when the initial forms of *all* the least-order cells vanish. If only some vanish, those cells drop out and the others give the limit at once (proposition 35). Every cell is non-negative, so their initial forms are non-negative at the arc coefficient and cannot cancel. So all of them vanish exactly when the initial form of their sum $P(A) = \sum_c f_c$, the event's total mass, vanishes. Degeneracy is therefore a property of one polynomial, $P(A)$, and resolving it is the classical Newton–Puiseux algorithm for that polynomial [18, 19]. The cells only decorate the result.

Setting. There are two parameters $(x, y) \to 0$ in the open quadrant. The cells $f_c \in \mathbb{R}[x, y]$ on $L \times A$ are non-negative there, and $P(A) \not\equiv 0$. Every semialgebraic arc into the corner can be written $y = \psi(x) = \sum_i c_i x^{e_i}$, a real Puiseux series with $0 < e_1 < e_2 < \cdots$ and $c_1 > 0$. This is possible

because x is strictly monotone along such an arc near 0. Iterated limits, which are not arcs, were treated in theorem 41.

Theorem 59 (recursive resolution of degenerate directions)**.** *(1)* The criterion. *Along ψ with leading term $c\,x^w$, the limit is read off at this level, as in proposition 35, unless w is the slope of a bounded edge of the Newton polygon of $P(A)$ and c is a root of that edge's polynomial $G_w(c) = \mathrm{in}_w P(A)(1, c)$. Hence the degenerate directions at a level are finitely many pairs (w, c^*).*

(2) The refinement step. *Let (w, c^*) be degenerate, with $w = p/q$ in lowest terms. The substitution $x = t^q$, $y = t^p(c^* + y_1)$ turns each cell into a polynomial $f'_c(t, y_1)$, non-negative for small $t > 0$ and small $|y_1|$ of either sign. The arcs with leading term c^*x^w correspond exactly to the tails $y_1 = \psi_1(t) = \psi(t^q)/t^p - c^*$, which tend to 0 or vanish identically. The conditioned state along ψ equals that of the refined cells along ψ_1. At refined levels, (1) applies with non-zero real roots of either sign. The exact branch $\psi_1 \equiv 0$ is a corner: its limit sends $t \to 0$ with $y_1 = 0$.*

(3) Termination. *Along any chain of refinements, the multiplicity of c^* as a root of the edge polynomial of $P(A)$ is non-increasing. A chain is infinite iff it follows an exact real Puiseux root $y = \varphi(x)$ of $P(A)$, and such a root is a common root of all the cells. Along it $P(A) \equiv 0$, so the arc lies in the wall. Every arc not lying in the wall is therefore resolved after finitely many refinements. Collapsing each infinite chain to its root, the refinement tree is finite: its nodes are the truncations of the finitely many real Puiseux roots of $P(A)$ with positive leading coefficient.*

(4) The limit set. *The limits along all semialgebraic arcs not in the wall form the finite union, over the nodes of the tree, of the non-degenerate limit maps $c \mapsto$ normalise$(\mathrm{in}\, f'(c))$ on each node's fan, together with the corner limits of the exact branches. Decisions are attached node by node, as in theorem 46.*

Proof. (1) By proposition 35, the leading coefficient of cell c along ψ is $g_c(c) = \mathrm{in}_w f_c(1, c) \geq 0$ if c attains the least order, and the cell is of higher order otherwise. Since $\sum_{\text{leading}} g_c = G_w$, with non-negative terms, $G_w(c) = 0$ iff every leading $g_c(c) = 0$. If w is not an edge slope, the least weight is attained at a single vertex, G_w is a monomial, and it has no non-zero root. (2) The substitution is exact. Non-negativity holds because $f'_c(t, y_1) = f_c(t^q, t^p(c^* + y_1))$ with $t^p(c^* + y_1) > 0$ for small y_1. An arc with leading term c^*x^w is $y = t^p(c^* + \psi_1(t))$. (3) This is Newton–Puiseux for the single polynomial $P(A)$. If c^* is an m-fold root of G_w, the refined $P(A)$ has a Newton polygon of height at most m in y_1, so the multiplicities of the next roots sum to at most m. A chain of constant multiplicity has bounded denominators and converges to a Puiseux root of $P(A)$ [18, 19]. Along a real arc each $f_c \geq 0$, so $P(A)(x, \varphi) \equiv 0$ forces every cell to vanish. A polynomial has finitely many Puiseux roots. (4) Combine (1)–(3). Checks: DG1 compares the recursion with direct expansion along 424 random arcs, generic and degenerate. DG2 checks the criterion, including 172 cases where only some leading cells vanish. DG3 and DG6 are the two examples below, exactly. DG4 checks, on 200 families with 302 tree nodes, that every node is a truncation of a root and that the recursion agrees with direct expansion along 996 arcs. DG5 checks that multiplicities are non-increasing and that every chain is finite. □

Example 60 (two refinements)**.** Source 1 has $f_1 = (y - x - x^2)^2$ and source 2 has $f_2 = x^5$. At level 0, $P(A)$ has an edge of slope 1 with edge polynomial $(c - 1)^2$, which is degenerate at $c^* = 1$ with multiplicity 2. Substituting $y = x(1 + y_1)$ gives $f_1 = t^2(y_1 - t)^2$: again slope 1, root 1, multiplicity 2. Substituting $y_1 = t(1 + y_2)$ gives $f_1 = t^4y_2^2$ against $f_2 = t^5$, which is non-degenerate. Back in the original parameters (fig. 13):

- an arc that leaves $y = x + x^2$ before order $\frac{5}{2}$ is explained by source 1;
- along $y = x + x^2 + bx^{5/2}$ the share of source 1 is $\frac{b^2}{1+b^2}$;
- an arc that leaves later, or follows $y = x + x^2$ exactly, is explained by source 2.

The limit sits two refinements deep, invisible to the first-level fan (DG3).

Example 61 (an infinite root, cut by a competitor)**.** Source 1 has $f_1 = (y^2 - x^2 - x^3)^2$, whose positive root is the infinite series $y = x\sqrt{1+x} = x + \frac{1}{2}x^2 - \frac{1}{8}x^3 + \frac{1}{16}x^4 - \cdots$. Alone, it gives an infinite chain of constant multiplicity 2: the arc runs inside the wall. With a competitor $f_2 = x^K$, the chain stops after exactly $\frac{K-2}{2} - 1$ refinements (2, 4 and 6 for $K = 8, 12, 16$), each following the series. At the cut exponent $e = \frac{K-2}{2}$, the share of source 1 is $4(c - a_e)^2/(4(c - a_e)^2 + 1)$, where c is the arc's coefficient and a_e the root's. The root's next coefficient appears as the zero of the limit map (fig. 13(c); DG6).

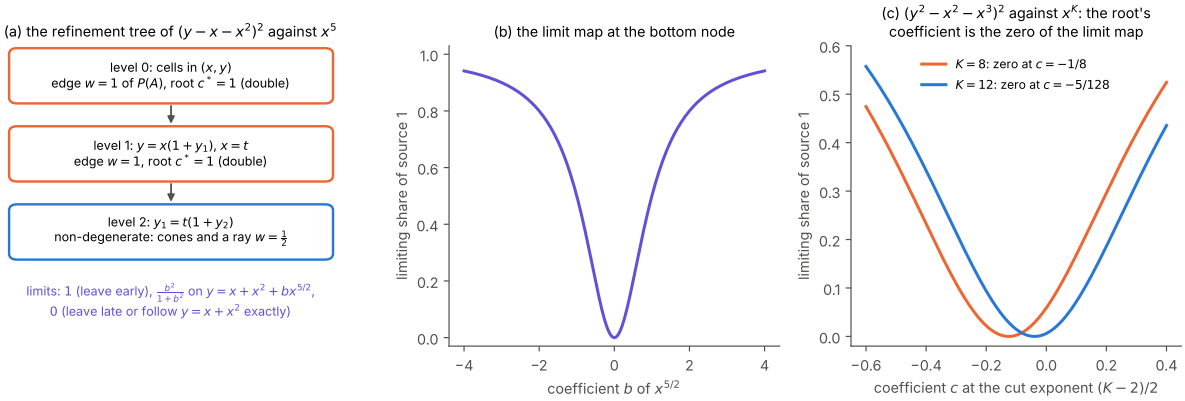


Figure 13: Step 4f. (a) The refinement tree of example 60. (b) The limit map at its bottom node. (c) Example 61: at the cut, the limit map vanishes exactly at the root's coefficient ($-\frac{1}{8}$ for $K = 8$, $-\frac{5}{128}$ for $K = 12$).

Scope. Two parameters. For $d > 2$ the same criterion holds: degeneracy is a property of $P(A)$ alone, since cells cannot cancel. The recursion then needs a resolution of $P(A)$ rather than Newton–Puiseux, for example by toric modifications along the degenerate faces. A termination bound there is open. Step 5 reads this recursion through the log map: degenerate directions are exactly where the log fails (theorem 63(5)).

23 Step 5: functions, duality and logs

Steps 1–4 built the covariant side: laws, kernels, the monad $\mathcal{D}$, the comonad W, the tangent structure T. Functions were present throughout, as events $\mathbf{1}_A$, utilities u_a and decision gaps, but they were never named as a side of their own. Neither were logarithms, though the order v of Step 3 is one. This step names both, adds the cotangent cone at the boundary, and collects what the layers say about what does not matter.

23.1 States and effects

A law p is a *state*; a function $u : X \to \mathbb{R}$ is an *effect*; $\langle u, p\rangle = \sum_x u(x)p(x)$ is their pairing. A kernel $K : X \to \mathcal{D}Y$ pushes states forward, $p \mapsto pK$, and pulls effects back, $u \mapsto Ku$ with $(Ku)(x) = \sum_y K(x,y)u(y)$. This is the state-and-effect picture of categorical probability [20].

structure	variance	where
$\mathcal{D}$, μ (laws, pooling); W, λ (provenance)	covariant	Step 1
T^k (feasible directions, Weil points)	covariant	Step 2
a kernel acting on states / on effects	covariant / contravariant	proposition 62(1)
marginalisation $M_S = \mathcal{D}\pi_S$	covariant in π_S, contravariant in the window $S \subseteq T$	proposition 6
window tables over a cover; the L^2 presheaf of conditional expectations	contravariant (presheaves)	covers paper; main paper, proposition 62(6)
the facewise presheaf $\mathcal{O}$	contravariant on the face poset	definition 49
parameterised families $P \mapsto \mathrm{Hom}(P, \Delta_{L\times X})$	contravariant in the parameters	definition 44
events, utilities, decision gaps	effects (contravariant)	proposition 62
the order fan (weights w are functionals on exponents)	dual to the Newton polyhedron	theorem 63(3)
the cotangent cone C_x^*	contravariant	proposition 64

Proposition 62 (conditioning is an update by an effect)**.** *(1) $\langle u, pK\rangle = \langle Ku, p\rangle$, and pulling back is contravariant: $(KL)u = K(Lu)$. Marginalisation pulls effects back along the projection, and forgetting provenance pulls u back to $u \circ \epsilon$.*

(2) For an effect $q \geq 0$ with $\langle q, p\rangle > 0$, the update *is $C_q(p) = q\,p/\langle q, p\rangle$. Then $C_A = C_{\mathbf{1}_A}$, updates compose by products, $C_r \circ C_q = C_{qr}$, and expectations update as $\langle u, C_q p\rangle = \langle qu, p\rangle/\langle q, p\rangle$.*

(3) The contextual Bayes decomposition holds for every effect. Updating the labelled law by $\epsilon^* q$ gives rows $C_q(p_l)$ and Bayes weights $\pi_l^q = \pi_l\langle q, p_l\rangle/\langle q, P\rangle$. Theorem 11 is the case $q = \mathbf{1}_A$.

(4) The Bayesian inverse is an update: $K^\dagger(\cdot \mid y) = C_{K\mathbf{1}_y}(p)$, the update by the point effect $\mathbf{1}_y$ pulled back along K.

(5) Pairing is natural for Para_L*: for a reparametrisation r, $\langle u, \Lambda \circ r\rangle = \langle u, \Lambda\rangle \circ r$.*

(6) The L^2 presheaf. *For a law p, the conditional expectation $E_p[\,\cdot \mid X_S]$ is the p-adjoint of pulling back along π_S. It is idempotent and satisfies the tower property $E[E[u \mid X_T] \mid X_S] = E[u \mid X_S]$ for $S \subseteq T$. So marginalisation of states and conditional expectation of functions are the two faces of one restriction: the main paper's L^2 presheaf, read in the layers.*

Proof. (1) Exchange the sums. (2) $C_r(C_q p) = rqp/\langle r, qp\rangle$ and $\langle r, qp\rangle = \langle qr, p\rangle$; the expectation formula is the same computation. (3) Row l of the updated labelled law is $\pi_l q\,p_l/\langle q, P\rangle = \pi_l^q C_q(p_l)$. (4) $p(x)K(x,y)/\sum_{x'} p(x')K(x',y)$. (5) Pairing is pointwise in the parameter. (6) $\langle E[u \mid X_S], v \circ \pi_S\rangle_p = \langle u, v \circ \pi_S\rangle_p$ is the defining property; idempotence and the tower property follow. FD1–FD3 check all six exactly, in rational arithmetic. □

Updates compose by *products* of effects. Under the log map these become sums, which is why Bayes is linear in logarithms and pooling is not.

23.2 The log map and the tropical limit

For a scale $s \in (0,1)$, write $\mathrm{Log}_s(a) = \log a/\log s$. The order of Step 3 is its limit: $v(f) = \lim_{s\to 0} \mathrm{Log}_s f(s)$ for germs asymptotic to powers.

Theorem 63 (logs and the tropical limit)**.** *(1)* The m-side tropicalises. *For positive $a_1, \ldots, a_n$,*

$$\min_i \mathrm{Log}_s a_i - \frac{\log n}{|\log s|} \leq \mathrm{Log}_s\Big(\sum_i a_i\Big) \leq \min_i \mathrm{Log}_s a_i.$$

So pooling, marginalisation and every other mixture become min *in the limit (theorem 30). This is Maslov's dequantisation.*

(2) The e-side is exact. $\log C_q p = \log q + \log p - \log\langle q, p\rangle$ *at every scale; posterior log-odds add the prior log-odds and the log-likelihood ratio. On valuations, conditioning is tropical division by the tropical pairing:*

$$v(C_q p)(x) = v(q(x)) + v(p(x)) - \min_y \big(v(q(y)) + v(p(y))\big).$$

(3) The order fan in log coordinates. *Along* $\varepsilon = c\, s^w$, $\operatorname{Log}_s \varepsilon \to w$ *and* $\operatorname{Log}_s f(\varepsilon(s)) \to \operatorname{trop}_w f = \min_\alpha \langle w, \alpha\rangle$. *The function* $\operatorname{trop}_w P(A)$ *is concave and piecewise linear. Its domains of linearity are the vertex cones of the order fan, and its breaks are the walls: the positive part of the tropical hypersurface of* $P(A)$ *[21]. A monomial reparametrisation acts linearly on log coordinates.*

(4) The m/e split. *Arithmetic mixtures (pooling: the m-flat families of information geometry) have valuation* $\min(v(p), v(q))$. *Geometric mixtures* $p^{1-t}q^t/Z$ *(e-flat families) have valuation* $(1-t)v(p) + t\, v(q)$ *minus its minimum, which is linear in* t. *Products of laws add valuations exactly, at every scale. So, in the limit, the dually flat pair of the main paper [22] becomes a pair of geometries: tropical-linear on the m-side and classically linear on the e-side.*

(5) Where the log fails. *The only operation without a log image is subtraction.* $v(a-b)$ *is not determined by* $v(a)$ *and* $v(b)$ *when they are equal. The degenerate directions of Step 4f and the contrasts that leave* $\mathcal{M}_k$ *in Step 4e are exactly the places where cancellation occurs.*

Proof. (1) $\max_i a_i \le \sum a_i \le n \max_i a_i$; divide the logarithms by $\log s < 0$. (2) Take logarithms; the valuation identity is theorem 30 applied to $\langle q, p\rangle$. (3) $\log \varepsilon_i = w_i \log s + \log c_i$. For the germs, apply (1). A minimum of linear forms is concave and piecewise linear, with one domain per vertex. (4) Exponents of monomial germs combine linearly under geometric mixing; products add exponents. (5) $s - (s - s^2) = s^2$. Checks: FD4 tests (1) on 500 random sums (largest ratio to the bound 0.90) and (2) exactly on germs. FD5 tests (3) by evaluation at scales $10^{-20}, 10^{-40}, 10^{-80}$ in log space, and the domain count. FD6 tests (4) exactly. □

23.3 The cotangent cone at the boundary

Proposition 64 (the cotangent cone)**.** *Let* $C_x^* = \{g : \langle g, u\rangle \ge 0$ *for all* $u \in C_x\}$, *with covectors taken modulo constants, and let* $N_x = -C_x^*$ *be the normal cone.*

(1) $g \in C_x^*$ *iff* g *is constant, say equal to* γ, *on the support of* x, *and* $g \ge \gamma$ *on its empty cells. At a relative-interior point* $C_x^* = 0$. *At the boundary it is a cone with one ray for each empty cell.*

(2) Contravariance. *Every smooth map* f *of simplices sends* C_x *into* $C_{f(x)}$, *so* $\mathcal{M}_1$ *contains them all. Hence* $Df(x)^{\mathsf{T}}$ *maps* $C_{f(x)}^*$ *into* C_x^*, *and* T^* *is a contravariant functor on dual cones.*

(3) Robust ties. *Let a linear decision tie at* x, $\langle u_a - u_b, x\rangle = 0$. *Then act* a *stays weakly preferred after every first-order injection iff* $u_a - u_b \in C_x^*$.

(4) The decision fan as a sign pattern. *Along an arc, the comparison of* a *and* b *is the sign of the germ* $\langle u_a - u_b, \Phi(\varepsilon)\rangle$. *Its numerator* $N_{ab} = \sum_c (u_a - u_b)(c) f_c$ *is a* signed *polynomial. On a cone of the fan, the decision walls are the positive roots of* N_{ab}*'s edge polynomial, and they*

are resolved by theorem 59 applied to N_{ab}. Unlike $P(A)$, N_{ab} can cancel, so decision walls appear in directions where the conditioned state itself is not degenerate.

Proof. (1) Pairing with the lineality directions $e_i - e_s$ forces g to be constant on the support. Pairing with the rays $e_z - e_s$ gives $g_z \geq \gamma$. (2) If $f(\Delta) \subseteq \Delta'$, then $x + tu \in \Delta$ gives $f(x + tu) \in \Delta'$, so $Df(x)u \in C_{f(x)}$. Then $\langle Df^{\mathsf{T}} g', u\rangle = \langle g', Df\, u\rangle \geq 0$. (3) The first-order change of the gap along u is $\langle u_a - u_b, u\rangle$. (4) The denominator $P(A)$ is positive; then apply theorem 46. Checks: FD7 pulls 6000 dual covectors back through stochastic maps and conditioning at boundary points, and tests (3) against 200 sampled injections per case. FD8 locates 9 walls at the predicted roots and checks that the act flips across each. □

23.4 What does not matter

The lab asked which parts of the data, the parameters and the model can be ignored for a decision [1, Section "What does not matter"]. In the language of this step, an exclusion is an annihilator, and the layers give five such statements.

Proposition 65 (exclusion principles)**.** *Let K be the window map of a cover and $\rho = (u_a - u_b)_{a,b}$ the utility gaps.*

(1) Duality. *The data directions that do not matter are* $\ker K \cap \ker \rho$. *Dually, the utility gaps that are functions of the window tables,* $\operatorname{span} \rho \cap \operatorname{im} K^{\mathsf{T}}$, *are pinned by the data and do not matter for settling. The decision-relevant dimension of [1, Proposition "the decision-relevant quotient"] is*

$$\operatorname{rank}(\rho|_{\ker K}) = \dim \operatorname{span} \rho - \dim(\operatorname{span} \rho \cap \operatorname{im} K^{\mathsf{T}}).$$

(2) Conditioning. *For a decision taken after observing A, the following do not matter (propositions 19 and 26):*

- *the data off A;*
- *the total mass of A;*
- *the provenance off A;*
- *the utilities off A.*

(3) At the wall. *Along a cone of the order fan, every coefficient off the leading face of N is irrelevant to the limit. On a vertex cone only the vertex's coefficients matter. Along a degenerate direction, only the coefficients at the node of theorem 59 that resolves it matter.*

(4) Feasibility. *Subtraction-free operations never matter for feasibility at any order (theorem 57); only contrasts do.*

(5) Statistics. *In the invisible cone of theorem 53 an adversary does not matter: its impact vanishes. On the wall side of theorem 55, standard errors stop mattering and counts are the right measure.*

Proof. (1) $\ker K = (\operatorname{im} K^{\mathsf{T}})^{\perp}$, so restricting ρ to $\ker K$ kills exactly its part in $\operatorname{im} K^{\mathsf{T}}$. (2) $\langle u, C_A p\rangle$ depends on u only through $u|_A$, and on p only through $p|_A$ up to scale. (3) The limit is the normalised initial forms (proposition 35, theorem 59). (4) and (5) restate the cited results. FD9 checks (1) on 40 covers, 14 of them with gaps pinned by a window, and checks (2) and (3) exactly. □

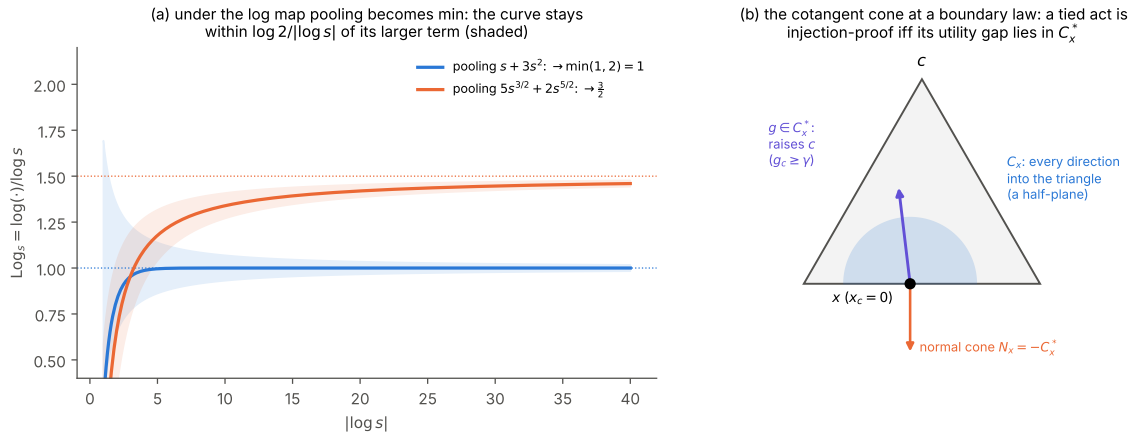


Figure 14: Step 5. (a) Under the log map, a pooled sum stays within $\log n/|\log s|$ of its leading term and tends to the minimum of the orders (theorem 63(1)). (b) At a boundary law the feasible directions form a half-plane. Covectors in the dual cone C_x^* cannot be lowered by any injection, and a tied act is injection-proof iff its utility gap lies there (proposition 64(3)).

24 Checks and scope

Step 1's checks are in `engine/check_context.py` (module `engine/pce/context.py`; 13/13, about a second). DL1–DL10, DL12 and DL13 use exact rational arithmetic; DL11 uses floating point. Step 2's checks are in `engine/check_tangent_layers.py` (module `engine/pce/tangent_layers.py`; 9/9, about 25 seconds) and `engine/check_jumps.py` (module `engine/pce/jumps.py`; 6/6, about 2 seconds; SJ1–SJ3, SJ5 and SJ6 exact). Step 3's are in `engine/check_orders.py` (module `engine/pce/orders.py`; 7/7, about 2 seconds; OR1–OR3, OR6 and OR7 exact). Step 4a–4b's are in `engine/check_fans.py` (module `engine/pce/fans.py`; 7/7, about two minutes; all exact except the nested numerics of MP5 and the Weil arithmetic of MP6). Step 4c's are in `engine/check_para.py` (module `engine/pce/para.py`; 6/6, under a minute; exact except the numerical comparison in PR3). Step 4d's are in `engine/check_statfan.py` (module `engine/pce/statfan.py`; 7/7, about a minute; ST1–ST3 exact or Weil, ST4 floating point, ST5–ST7 Monte Carlo with fixed seeds). The Poisson band and Step 4e are in `engine/check_graded.py` (modules `engine/pce/poissonband.py` and `engine/pce/graded.py`; 11/11, about two minutes). Step 4f's are in `engine/check_degen.py` (module `engine/pce/degen.py`; 6/6, exact). Step 5's are in `engine/check_effects.py` (module `engine/pce/effects.py`; 9/9, under a minute). TL8 is exact; the others use floating point with the Weil-algebra arithmetic of the tangent companion. The figures come from `context/fig_layers.py`.

DL1–DL4	the four axioms of the distributive law (proposition 3), exactly, on 50 random elements each
DL5	the two-sided Kleisli category: explicit composite, associativity, identity
DL6	marginalisation: iterated, affine, commutes with forgetting provenance
DL7	conditioning: partial domain, total probability, iteration
DL8	the conditioning–marginalisation square: commutes for cylinders, fails for a non-cylinder event
DL9	a known source stays known; the contextual Bayes decomposition; the old weights are wrong
DL10	forgetting provenance: commutes with windows, not injective, labelled windows differ
DL11	$D^2M = 0$; DC_A, D^2C_A against finite differences; $\|DC_A\|\,p(A)$ constant as $p(A) \to 0$
DL12	no continuous extension of C_A to the wall (proposition 10)
DL13	the worked examples: $\frac{8}{23}, \frac{15}{23}$; $\frac{9}{23}$ against $\frac{63}{160}$; $\frac{9}{19}$ against $\frac{23}{40}$
TL1	T commutes with W: $T^2\lambda$ lands in $T^2\Delta_{L\times X}$; counit and comultiplication on T^2 points (proposition 15)
TL2	the source face: tangent directions stay on the row; every other feasible direction is a face direction plus an injection (proposition 16)
TL3	forgetting maps cones onto cones; injection and drift have the same image; relabelling is invisible
TL4	marginalisation: explicit lift onto, T^2 into T^2, kernel dimension $\|X\| - \|X_S\|$, $D^2 = 0$ (proposition 17)
TL5	conditioning: lift onto, kernel dimension $\|X\| - \|A\|$, second-order criterion and 900 direct T^2 points, pole; a negative control fails both tests (proposition 19)
TL6	pooling: first order into the cone, the criterion at corners including a weight-0 source, 1200 direct T^2 points, $D^3 = 0$
TL7	the differentiated Bayes decomposition (proposition 23)
TL8	example 24 exactly: $\frac{9}{23}$, $-\frac{16}{529}$, $\frac{384}{12167}$, $\dot\pi_1^A = \frac{640}{529}$, and the reweighting split
TL9	theorem 20: explicit tensor decompositions ($k \le 4$), closure under products, pooling at order 3
SJ1	theorem 25, exactly, on 291 families with zero weights and zero likelihoods
SJ2	proposition 26(1)–(2), with the coinciding-conditionals example
SJ3	proposition 26(3)–(4): explicit lifts, fibre dimensions, any posterior provenance, identifiability only lost
SJ4	theorem 27(1)–(2): the rate bound on random paths, the provenance bound, an oscillating extinction
SJ5	theorem 27(3)–(5), exactly: leading-jet limits, the smooth example, a removable passage, a kink
SJ6	example 28, exactly: width 4η, slope $-1/(4\eta)$, the replacement bound
OR1	theorem 30: order is a homomorphism to $(\min, +)$ on non-negative germs; cancellation without positivity
OR2	corollary 31: every row of the order calculus, exactly, on 200 labelled paths
OR3	theorem 32(1)–(2): limits, survivors, extinction rates, the error exponent δ
OR4	theorem 32(3): analytic passages agree; the C^∞ flat jump
OR5	theorem 32(4): Hardy (limit) against oscillating (none), in log space
OR6	proposition 33: the k-jet sees the transition iff $k \ge m$; the Weil coefficients $\|S\|!\,c_{\|S\|}$
OR7	example 34, exactly: width $2(2\eta)^{1/m}$ for $m = 1, 2, 3$
MP1	proposition 35: arc limits are normalised initial forms, exactly, 300 families
MP2	theorem 36(1): the leading faces are the vertices and bounded edges of N
MP3	theorem 36(3)–(4): the componentwise-minimum criterion for monomial cells
MP4	examples 37 to 40, exactly
MP5	theorem 41: iterated limits are lexicographic vertices; orders disagree; corners agree while the middle differs
MP6	proposition 42: Weil points in T^d, the lemma's first term, the flip
MP7	example 39: lines agree, parabolas do not
PR1	proposition 45(1): Para composition, identities, whiskering, exactly
PR2	proposition 45(2)–(3): layer operations commute with reparametrisation; the conditioning domain pulls back
PR3	the germ field: valuations are tropical orders, the standard part is the arc limit

Scope. Finite sets and discrete, known sources. Step 2 settles first order for every operation, order two for conditioning (every order, by Step 4e), and every order for forgetting, marginalisation, the source embedding and pooling. It classifies where conditioned probabilities can jump on finite sets. Step 3 gives the order calculus of transitions along one-parameter paths, and Step 4a–4b extends it to several parameters for polynomial cells (a full existence criterion for general polynomial cells is open). Step 4c attaches decisions to the fan, and Step 4d derives Theorem J's band from the cross term under J's Gaussian model (theorem 53). Step 4d adds an exact band on the Poisson ray, and Step 4e shows that every layer operation, conditioning included, keeps feasibility at every order. Step 4f resolves degenerate directions recursively for two parameters. Open: termination bounds for more than two parameters, and bands for decoration-dependent decisions near walls.

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A Symbols across the companions

The papers share one notation. The column "where" says which paper uses a symbol (L the lab, C the covers companion, T the tangent companion, Y the layers companion). Where a letter means two things, both meanings are listed and the clash is flagged; context always decides, and no paper uses both meanings in the same section.

symbol	meaning	where	note
Laws and the two-type model			
P, Q	the true law; an injected law	L, T	P is also C's parity matrix
m, n	number of binary variables; sample size (draws per window)	all	
$w = w_1$, w_2	weights of type 1 and type 2, $w_1 + w_2 = 1$	all	
$a_1, a_2 \in [0,1]^m$	success (harm) probabilities of the two types	all	
μ	means, $\mu = w_1 a_1 + w_2 a_2$	all	
v, v_S, v_T	latent factor $v = \sqrt{w_1 w_2}\,(a_1 - a_2)$; $v_S = \prod_{i \in S} v_i$; v_T for a triple T	all	
t, r	$t = \sqrt{w_2/w_1}$; $r = t - 1/t = (w_2 - w_1)/\sqrt{w_1 w_2}$, the one unseen coordinate	all	

symbol	meaning	where	note
m_S, c_{ij}	central moment of the window S; pair moment $c_{ij} = m_{\{i,j\}} = v_i v_j$	all	
$s_k(r) = M_k$	$m_S = v_S\, s_{\lvert S\rvert}(r)$; s_k are Fibonacci polynomials, the moments $E\tau^k$ of the standardised latent	L, C	
χ	the move $\chi(x) = \prod_i (2x_i - 1)$: laws with the same pair tables are $P + s\chi$	L, T	
$R = [r_-, r_+]$, $t_\pm$	the hiding segment and its walls (positivity of the explanation)	L, T	
Covers and their algebra			
G, E, G^*	graph of observed pairs; its edges; the non-zero graph	C	
$b(G)$, c^*	number of bipartite components; components of G^*	C	
B, A	unsigned incidence matrix; extended incidence matrix (pairs and triples)	C, L	
I_A, I_B	toric ideal of A (of the graph)	C, L	
K, F	a cover as a simplicial complex; a face (window)	C	K is also a torus and, in T, a cone
φ	moment map $(v, r) \mapsto (m_F)_F$	C	
δ, H^0_{inc}, H^1_{inc}	multiplicative incidence complex $(\delta v)_{ij} = v_i v_j$ on G^*; its kernel and cokernel	C	not a Čech complex
$H^1_+(\Gamma; \mathbb{R}^\times)$	L's name for H^1_{inc} on a connected pair graph (Theorem M(7))	L	same group as C's H^1_{inc}
$q(G^*)$	number of non-bipartite components of G^*	C	
λ_k, y, x_k	bipartite scale; $y = \log t$, $x_k = \log \lambda_k$	C	λ^* in L is the gluable fraction
Π_σ, σ	the sheet polytope for the component signs σ	C, T	
G_2, P	sign group $\ker P^\top$ over $\mathbb{F}_2$; parity matrix	C	
ℓ_i	lobes at a vertex i	C	T's ℓ is the vertical lift
τ (in C)	standardised latent, $\tau = t$ or $-1/t$	C	L's τ is the tempo constant
τ_z, u_i (in C)	whitened atoms and frame vectors of three types, $a_{z,i} - \mu_i = u_i \cdot \tau_z$	C	$O(2)$ acts on both
π_{UV}, g_{PQ}	label transition in S_k; frame transition in $O(2)$	C	
Decisions and probes			
u_a, U_a, W_a	utility of act a; its expected value; its worst value on an interval	L, T	
u_0, $\sigma(I)$	requirement; the set rule on an evidence interval I	L	σ in C is a sign vector
β_a, c_a	slope of U_a in r; the Walsh coefficient $\sum_x u_a(x)\chi(x)$, $\beta_a = v_T c_a$	L	c in T is the flip
d, n^*	settling radius; median settling size $(z\tau/d)^2$	L	$d = a_1 - a_2$ in L's setup
τ (in L), τ_c, τ_m	tempo constant (asymptotic sd of $\sqrt{n}\,\hat{r}$); its conditional and plug-in versions	L	
ψ	three-way log odds ratio $\sum_x \chi(x) \log p_x$	L	
ε, ε^*, $\varepsilon_{\mathrm{det}}$	injected share; the share that flips the act; the share that is detected	L, T	
$A \boxplus B$, $A^\perp$	partial sum $A \cup B$ of disjoint events; orthocomplement $X \setminus A$ (Roumen's effect-algebra terms)	L	$\boxplus$, not $\oplus$ (direct sum); $(L \bmod 2)^\perp$ in C is an annihilator over $\mathbb{F}_2$

symbol	meaning	where	note
test, state	a window's partition of X into cylinder events; a map to $[0,1]$ preserving 1 and $\boxplus$ (a law)	L, C	elsewhere "test" is a statistical test and w in a string diagram is a state
$\mathcal{D}$, η, μ	finite distribution monad; point mass; flattening (pooling)	Y, L	μ is also the mean in L
$W = L \times (-)$, ϵ, δ^W, λ	provenance comonad (sources, not measurement contexts); forget the source; copy it; distributive law $(l,p) \mapsto \delta_l \otimes p$	Y	λ is also a scale in C
M_S, C_A, π_l^A	marginalisation onto S; conditioning on A (partial); Bayes-updated source weights	Y	C_X in L conditions on an observable
$\mathfrak{C}_A$, L_A, $\mathcal{S}_J(A)$	labelled conditioning $C_{L\times A}$; surviving sources; strata by surviving set	Y	
v, m, m_l, $\mathcal{H}$	order of a germ (valuation); order of $P(A)$ and of source l's mass on A at a transition; a Hardy field	Y	m is also the number of variables elsewhere
trop_w, in_w, N	tropical order and initial form along weights w; Newton polyhedron of the cells on $L \times A$	Y	
Para_L, $\mathbb{K}$, D_{ab}	parameterised source-indexed kernels (2-cells: reparametrisations); real Puiseux series; comparison germ of acts a, b	Y	
$(\Sigma_N, \mathcal{O})$, ℓ_F	probability-decorated order fan (geometric fan, facewise presheaf of initial forms); limit map of a face	Y	
$\mathcal{M}_k$, B_{pois}, t_b	graded class of maps keeping feasibility at order k; exact Poisson band; blind-spot total	Y	$\mathcal{M}$ unrelated to M_S
$\langle u,p\rangle$, C_q, C_x^*, Log_s	pairing of an effect and a state; update by an effect q; cotangent (dual) cone; $\log(\cdot)/\log s$	Y	C_q extends $C_A = C_{\mathbf{1}_A}$
μ_n, J, $\dot{\pi}_l^A$	pooling n sources; the labelled joint $\sum_i w_i\, \lambda(i, p_i)$; derivative of a Bayes weight	Y	
$K_p^\dagger$, C_X, Rect_X	Bayesian inverse of K at p; conditioning operator; rectangular hull along X	L	$\dagger$ is not an adjoint here
ρ, R, $V[X]$	utility differences; Bayes risk $\min_a E_P \ell_a$; value of observing X, $R - X.R$	L	R is also the hiding segment and a ring
λ^*	largest gluable mass of a family of windows	L, T	in T's layers also the scale of c_{01} where the triangle's explanations end (0.3115)
Tangent structure			
T, T^k, T_2	the tangent functor, its iterates, the second-order pullback	T, L	T is also a triple in L and C
p, 0, $+$, ℓ, c	projection, zero, addition, vertical lift, canonical flip	T	
W_k, e_i	Weil algebra $\mathbb{R}[e_1,\dots,e_k]/(e_i^2)$ and its generators	T	
C_x, $C_{x,u}$	feasible cone at x; the cone after moving infinitesimally along u	T, C	
$\geq_{\mathrm{lex}}$	lexicographic order (first non-zero entry positive)	T	
J, $[H(u,v)]$	Jacobian of the tetrads; the completion obstruction in coker J	T	
II, F, h, q	second fundamental form of a decision boundary; decision functional; chord polynomials	T	F is a face in C

symbol	meaning	where	note
$\bar{f}$, e_F	restriction of a partial map; restriction idempotent of a face	T	

B Proof-status register

Each main statement of the companions, with one of four statuses. *Proved*: a complete argument is in the paper. *Verified*: exact or numerical checks support it, with no general proof. *Conjectured*: a plausible general statement, with evidence but no proof. *Open*: neither an argument nor a counterexample. A statement can carry two statuses for two parts, and the scope (types, cover, number of variables) is part of the statement. L, C, T, Y as in appendix A.

	statement	status	scope and evidence
L	A: hiding segment; B: tempos; C: value of a probe	proved	two types, pair cover (+ triples), $m \geq 3$
L	D(1)–(3): existence, uniqueness, invariance of the set rule	proved	two types, pair cover
L	D(4): settling at $n^* = (z\tau/d)^2$; never on the stratum	proved (asymptotic)	$m = 3$, affine utilities; Study 11b verified
L	D(4) for $m \geq 4$ (quadratic utilities)	open	
L	E slopes, vertex attack; J source bands	proved	any law; J in the Gaussian limit
L	F hidden channel; G stealth with equations; H one geometry	proved	F at $m = 3$ (part 4 any m); G at $m = 4$; H at $m = 3, 4$
L	K lifting; L one picture; N assembly	proved	as in the scope table
L	M descent: sign and modulus classes	proved	any pair graph; acyclicity verified for $m = 3, 4$
L	rectangularity as gluing: sharpness	conjectured	finite trees
L	ring of tables: fibre, tropical limit, Gröbner form, $\tau_c \leq \tau_m$	proved	three binary variables (checks EH1–EH7)
L	exact probe settles before the plug-in probe	false in general	Study 11c, E2 failed on D5
L	Bayesian inversion: coupling swap, involutive a.s., reverses composition, partial (domain ends at a positivity wall); C_X built from it	proved	IV1–IV4
L	defect sandwich $0 \leq D_X(F) \leq \text{ante}(F) - \text{ante}(\text{Rect}_X F)$; bound attained in some cases	proved	IV5, IV6
L	inversion and the one-sided geometry: affine inside the data, linear-fractional, preserves feasibility, no pooling flips, pole $1/q(y)$	proved	IV7–IV10
L	comparison natural transformation between conditioning and aggregation, invertible iff rectangular	open	
L	wall-crossing formula for the value of a probe (piecewise linear; kinks on decision and conditional walls)	proved	RL9

	statement	status	scope and evidence
L	decision walls: joint chamber complex with positivity walls; decision-identifiable region; robust wall-crossing	open	a programme
L	decision-relevant quotient: gauge, $\rho(F(O))$, relevant rank, redundant windows	proved	RL1–RL4
L	decision value $V = R - X.R$ is a Vigneaux 1-cocycle; latent-class families adapted (curved); irrelevance iff common optimal act	proved	RL5, RL6
L	rectangular beliefs: ex-ante = updated (Epstein–Schneider); the hiding segment is not rectangular, updating loses value	proved (cited) / verified	RL7, RL8
L	magnesium before ISIS-4: decision-settled only at $h = 0$ (M3 failed, margin 0.12 s.e.)	verified (Study 14)	published trials; band identity only
C	factorisation on any pair graph (zero, sign and modulus classes)	proved	any graph
C	hiding polytopes; dimension $b(G)+1$; 2^{c^*-1} sheets	proved	$c^* \geq 1$
C	algebra, reals, positivity (bridge lemma)	proved	pairs and triples
C	toric relations; syzygies = dependencies among tests	proved	Wald reading needs the usual regularity
C	Fibonacci gluing and the Hankel relation	proved	any window sizes
C	Betti tables of the computed covers	verified	Macaulay2 = Singular; graded-piece generators for all
C	two largest covers: generators 71, 159; $\beta_2 = 481$ (two tetrahedra); higher Betti numbers	partly computed	$\beta_{\geq 3}$, $\beta_{\geq 2}$ exceed 8 GB
C	collinear k types: moment space, M_3 range unchanged	proved	any k
C	three types on five variables: local identifiability from two windows	verified	ranks at generic points
C	global S_k descent through a shared overlap (Kruskal; Jennrich construction)	proved	any k, windows sharing O; TH1, TH2, TH7
C	three types, $m = 5$: generic global identifiability up to S_3 under (G1), (G2)	proved	two windows meeting in a triple
C	whole-overlap agreement: not identifiable (3-parameter family); equal weights not needed	proved	TH4
C	ranks 15 (two overlap variables agree), 16 (collinear private points)	verified	TH5
C	every overlap variable shared by a different pair: rank 17, global uniqueness	open	no pivot meets Kruskal
C	label holonomy: trivial for a true k-type law on any cover; trivial holonomy glues; a non-trivial 3-cycle example	proved	identified nodes, distinct overlap profiles; TH8–TH10
C	real signs, invisible classes via the Smith form	proved	any cover with pairs and triples
C	stratified descent: support; sign $\oplus$ modulus as one class in H^1_{inc}; torsor under H^0_{inc}	proved	any pair graph, every stratum
C	positivity: Farkas; negative cycles; two components suffice	proved	every sheet (checks LV5, LV8)

	statement	status	scope and evidence
C	strata: latent rank, degree rule, $r = 0$, the merge	proved	any cover
C	lobe rule on pair graphs: exactly $\max(0, \ell_i - 1)$ lost	proved (Lemma "embedding dimension at a face")	any pair graph; CS4, CS9
C	lobe rule for covers with windows: loss = $\operatorname{rank} A_{K-i} + \mu_i - \operatorname{rank} A$	proved	pairs and triples; CS9
C	frames: pairs = off-diagonal rank-2 Gram; hidden circle of laws at fixed frame	proved	three types (FR1, FR9)
C	$O(2)$ frame cocycle on rigid patches; trivial for true data; twists by any angle; window layer induced by S_3	proved	spanning overlaps (FR2–FR6)
C	rank-one holonomy = sign around the lift; two-type label holonomy = sign class	proved	generation hypothesis on cycles (FR7, FR8)
C	rank-2 rigidity of graphs (K_5, K_6 rigid; K_4, $K_{3,3}$, Petersen not)	verified	FR4
C	complete graphs, $n \geq 5$: u generically identified up to $O(2)$ (Anderson–Rubin)	proved (cited)	FR5 agrees
C	realisations form a prestack, not a stack; twisted realisations classified by $H^1(\text{nerve}; S_k)$; law layer separate; $O(2)$ on the Gaussian shadow	proved	identified nodes, distinct profiles
C	coincident profiles (double cosets); global rank-2 rigidity of incomplete graphs; flexes of non-rigid parts	open	
C	Phases IV–V: (F) torsor, local (P) rank, global (P) by G_2 then positivity walls, (L) = (P) with three loading variables; strict ladder	proved	ID1–ID6
C	closed description of the positivity region for large G_2	open	
T	cone structure: (T1)–(T5); (T6) iff relative interior	proved	polyhedra with affine maps; verified on Δ_3, Δ_4
T	jets on the tetrad scheme; obstruction $[H(u, v)]$	proved	$m = 4$
T	second-order morphisms at corners	proved	smooth maps of simplices
T	order-k condition at corners: necessity	proved	every k
T	order-k sufficiency (conjecture); second-order subcategory not closed under T^3	conjectured / proved	$k = 3$ computed (O31–O33); counterexample $X_1(X_1 - X_2)^2$
T	pooling paradox from curvature	proved (local)	sufficient construction near a boundary point
T	layers; resolutions; multi-maps	proved	resolutions are chosen real ones
T	restriction category (R1–R4, joins), T a restriction functor	proved	verified on 600 laws
T	full differential-restriction axioms	open	
T	tangent cone of a sheet; decisions on a sheet	proved	
T	pooling along a chord	proved	$h \not\equiv 0$; exact root isolation

	statement	status	scope and evidence
Y	Step 1: $\lambda : W\mathcal{D} \Rightarrow \mathcal{D}W$ is a distributive law (four axioms); the two-sided Kleisli category of context-indexed kernels	proved	finite sets, discrete contexts; DL1–DL5
Y	marginalisation total and affine; conditioning partial and nonlinear; total probability; iterated conditioning	proved	DL6, DL7
Y	conditioning commutes with marginalisation for cylinder events (not in general)	proved	DL8
Y	contextual Bayes decomposition; forgetting provenance commutes with marginals and is not injective	proved	DL9, DL10, DL13
Y	derivatives of conditioning, $\lvert DC_A\rvert \sim 1/p(A)$; no continuous extension of C_A to its wall (categorical conditionals are arbitrary there)	proved	DL11, DL12
Y	Step 2: T commutes with W; each source is a face, and a direction leaves it exactly by injecting records under another label; injection and drift agree after forgetting	proved	TL1–TL3
Y	marginalisation and conditioning map feasible cones onto feasible cones (explicit lifts, kernels); conditioning's lift scales with $p(A)$	proved	TL4, TL5
Y	T^kH is generated by elementary tensors, so closed under products; products of laws and pooling are morphisms at every order	proved	TL6, TL9 (upgrades T's "checked" TG5 claim for products)
Y	differentiated contextual Bayes decomposition (reweighting + within-source); worked example $-\frac{16}{529}$	proved	TL7, TL8 (exact)
Y	strata and jumps: provenance support $\{\pi_l p_l(A) > 0\}$; weights preserved / pooling commutes criteria; conditioning maps forgetting-fibres onto fibres (identifiability only lost)	proved	SJ1–SJ3 (exact)
Y	regimes: rate $\leq 2/P(A)$ in the domain; wall limit = conditional of the leading jet; analytic passages removable; jumps need a kink, a non-analytic path or a jump of the law (finite sets)	proved	SJ4–SJ6
Y	hybrid dynamics with transition operators; regimes beyond finite sets	open	remark in Y
Y	Step 3: order is a semiring homomorphism to $(\min, +)$ on non-negative Hardy germs; order calculus of every layer operation	proved	OR1, OR2 (exact)
Y	hierarchy of transitions: least-order sources survive (weights $c_l/\sum c_j$), extinction orders $m_l - m$, one-sided data, flat approaches (limit exists in a Hardy field, not outside); order m = level of the tangent tower that sees the transition	proved	OR3–OR7

	statement	status	scope and evidence
Y	Step 4a: several parameters: arc limits = normalised initial forms; order fan = normal fan of the Newton polyhedron; limit set cone by cone; componentwise minimum suffices (and is necessary for monomial cells); degenerate directions need refinement	proved	MP1–MP4, MP7
Y	Step 4b: iterated limits = lexicographic vertices (corners of the fan); multilinear Weil points lie in T^d, the lexicographic lemma computes one corner and the flip the other	proved	MP5, MP6
Y	Step 4c: parameters as an object (Para over source-indexed kernels, 2-cells = reparametrisations); layer operations commute with reparametrisation; germ field (Puiseux) reading; decision fan: eventual acts along arcs, ties broken at higher order, transport along monomial reparametrisations	proved	PR1–PR6
Y	Step 4d: probability-decorated order fan (facewise presheaf of initial forms; one N, different limits); exact cross term of pooling; Theorem J as an inequality of orders on the statistical fan (band = cross-term coefficient on the exposure ray; vanishing share; no tilt) under J's Gaussian model	proved	ST1–ST5
Y	Gaussian cone / Poisson ray / empirical wall for conditional estimates	proved (Monte Carlo illustrated)	ST6, ST7
Y	exact band on the Poisson ray (conditional binomial test): monotone exposure, Gaussian limit = J's band, blind spot, wall side (adversary owns the rare event), under (S1), (S3)–(S5)	proved	PB1–PB5
Y	Step 4e: graded classes $\mathcal{M}_k$ (descending chain of subcategories, strict at orders 2 and 3); subtraction-free maps in $\mathcal{M}_\infty$, so every layer operation, conditioning included, keeps feasibility at every order	proved	GC1–GC6
Y	Step 4f: degenerate directions (two parameters): refine iff $P(A)$'s edge polynomial vanishes (cells cannot cancel); Newton–Puiseux refinement of $P(A)$; multiplicities non-increasing; infinite chains = exact common roots (in the wall); finite tree; limit set node by node	proved	DG1–DG6
Y	Step 5: state–effect duality (conditioning, Bayes weights, Bayesian inverse as updates by effects; Para naturality; L^2 presheaf as the effect side of marginalisation)	proved	FD1–FD3

	statement	status	scope and evidence
Y	logs: Maslov bound (pooling $\to$ min), Bayes exactly linear in logs, conditioning = tropical division on valuations, order fan = normal fan in log coordinates (tropical hypersurface of $P(A)$), m/e split; subtraction is where logs fail	proved	FD4–FD6
Y	cotangent cone C_x^* (contravariant), robust ties, decision walls = positive roots of the signed N_{ab}; exclusion principles (annihilator identity for the decision-relevant dimension, conditioning, wall, feasibility, statistics)	proved	FD7–FD9
Y	termination bounds for $d > 2$ parameters; bands for decoration-dependent decisions near walls	open	planned
Y	conditioning beyond order two in the layers	proved (Step 4e)	GC5